City of Buffalo

PROPOSAL FOR

Buffalo Police Shooting Range Facility
Architectural, Engineering & Environmental Services
Job# 1917

July 12, 2019

submitted by:

LiRo Engineers, Inc.
A LiRo Group Company

July 12, 2019

Mr. Stephen M. Buccilli, P.E.
City of Buffalo
65 Niagara Square
Room 616 City Hall
Buffalo, New York 14202

RE: Proposal for Professional Architecture, Engineering & Environmental Services for the Public Improvements for the Buffalo Police Shooting Range Facility

Mr. Buccilli,

LiRo Engineers, Inc. (LiRo) is pleased to submit our proposal to the City of Buffalo to provide Professional Architecture, Engineering & Environmental Services for the Public Improvements for the Buffalo Police Shooting Range Facility. As you find in reading our proposal, LiRo is uniquely qualified by having a depth of available experienced local staff with a breath of shooting range design experience.

Our range design team has extensive experience in developing indoor shooting ranges, including short-range pistol and long-range rifle fire as long as 100-yards. The engineering challenges associated with shooting ranges are different from the challenges for typical building design. We have successfully met these challenges to meet the needs of our clients. Our in-house LiRo Team includes range designers, acoustical modelers, HVAC engineers, civil engineers, and electrical engineers, all of whom live in Western New York and work out of our office on Delaware Avenue in Buffalo. In addition to our own in house shooting range experts, our team includes Foit Albert Associates, a firm that we have successfully teamed with on several of our range projects.

Over the last 15 years, the cost of shooting range construction has escalated stressing the abilities of many municipalities to balance available funding to meet the training needs of law enforcement officers. One reason for the cost escalation is that stringent design criteria is being applied to range designs across the United States. Through the course of design phasing, LiRo's design team will discern, by close coordination with the Buffalo Police Department, how the range will be utilized and will rely on appropriate operational rules such that the basis of range design does not result in an unnecessarily costly over design. In this manner, a range that is cost effective for the type of shooting that is practiced will be provided to the City, without sacrificing safety or environmental stewardship.

LiRo is focused on completing our projects to our clients' satisfaction. This includes maintaining schedules and providing cost effective results. Our project team has successfully managed and controlled projects and schedules through the aspects of project development including data gathering and environmental permitting; design and bid support; submittal review and construction management; and closeout.

As a leading New York State construction management firm, LiRo has a long history of efficiently managing construction projects and we bring that experience to bear in our designs. Our philosophy involves the integration of controls on cost, time and quality in meeting the client's concerns and requirements, and the development of a detailed construction plan. LiRo designs are based upon familiarity with the relevant engineering, architectural, and professional services associated with the project and integrated project cost and schedule controls. LiRo's capacity to integrate "in house" constructability review and value engineering services allows our design team to propose creative and thoughtful designs that can be efficiently implemented. Our designs will allow activities to be conducted in such a way as to minimize public disturbance, risk, and facility operations disruptions, and to ensure public safety in the vicinity of construction activities.

We trust that our submittal will show that LiRo is qualified, experienced, and ready to work with the City on this important project. Should you require any additional information or wish to discuss our qualifications in more detail, please contact me directly at 716-882-5476. ***We welcome the opportunity to meet with you to discuss our proposal in more detail prior to consultant selection.***

Sincerely,
LiRo Engineers, Inc.

[redacted]

Robert Kreuzer
Senior Vice President

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Related Firm Experience 1

LiRo has gained a reputation as the firm of choice for many of New York's most demanding and sophisticated clients. Our reputation has been earned through our consistent delivery of the highest-level professional understanding of the technical, regulatory, operational, budgetary and scheduling requirements of design and construction. The experience of the LiRo project team translates into reliable, buildable design and construction documents, and a cost effective, timely and qualitative project delivery that is well coordinated with the stakeholders.

LiRo has been working with the New York State Office of General Services the New York City Police Department, the New York City Department of Environmental Protection, New York City Police Department and Amherst Police Department to remediate firing range sites, and to design and renovate new outdoor and indoor shooting ranges. Over that time, our civil, structural, environmental, and MEP professionals have completed design of correctional facility ranges and designed renovations of police precinct ranges.

We have completed new/replacement firing range design, from program stage, through permitting, acoustic analysis and plan and specification preparation, of state-of-the art indoor ranges and outdoor range facilities. Through the course of design phasing, LiRo's design team has a track record of discerning, by close coordination with our Clients, how each range will be utilized, relying on appropriate operational rules such that the basis of range design does not result in an unnecessarily costly over design. In this manner, a range that is cost effective for the type of shooting that is practiced is delivered, while completely addressing health, safety, environmental and neighborhood stewardship.

LiRo will conduct the work from our office at 690 Delaware Avenue in Buffalo. Robert Kreuzer, LiRo's Buffalo office principal and Mike Lydle, the assigned Project Manager, have managed many projects within the City of Buffalo. Our firing range personnel and civil, mechanical, electrical, plumbing and environmental engineers with the range experience described below work out of our Delaware Avenue office providing the City with an unmatched combination of range design experience and local responsiveness. LiRo is familiar with the City's design and bid process.

In addition, LiRo has developed an excellent working relationship with Foit-Albert Associates who have successfully supported LiRo in providing Code Review and Architectural services for several of our range projects including the NYSOGS range projects and Amherst PD project. This strong effective relationship will continue in serving the City of Buffalo, Department of Public Works (DPW) and Buffalo Police Department (BPD) on this project.

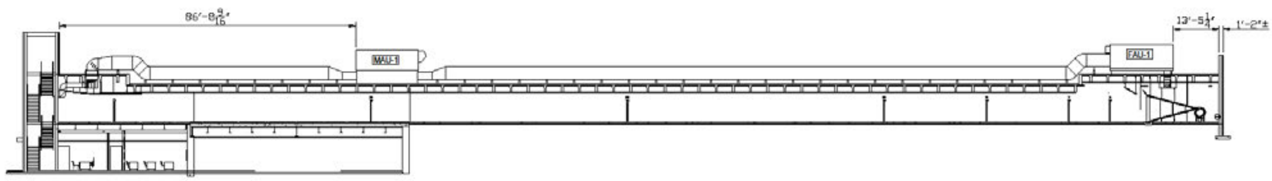
Presented below are descriptions of a few of the more recent relevant projects LiRo has provided:

New York State Office of General Services
Firing Range Design and Rehabilitation, Statewide

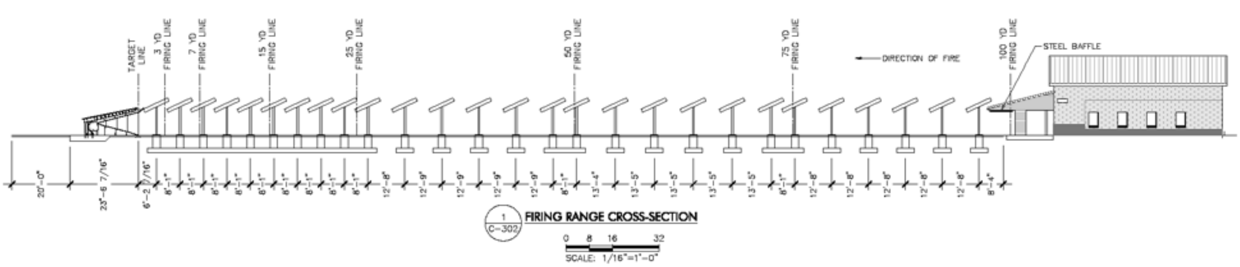
LiRo Engineers, Inc. was awarded a contract for the design firing ranges – statewide. Thus far, three projects have been awarded for design of one outdoor and two indoor correctional facility ranges.

Programming/System Analysis - Each design was developed by thoroughly reviewing each facility's varied training, access, classroom, communication, utility integration, parking and other requirements during multiple site visits, meetings and follow-up correspondence. Multiple-design options were evaluated. The designs meet ballistic containment, ventilation, acoustical dampening, OSHA, ADA, SEQR, various environmental and Code requirements.

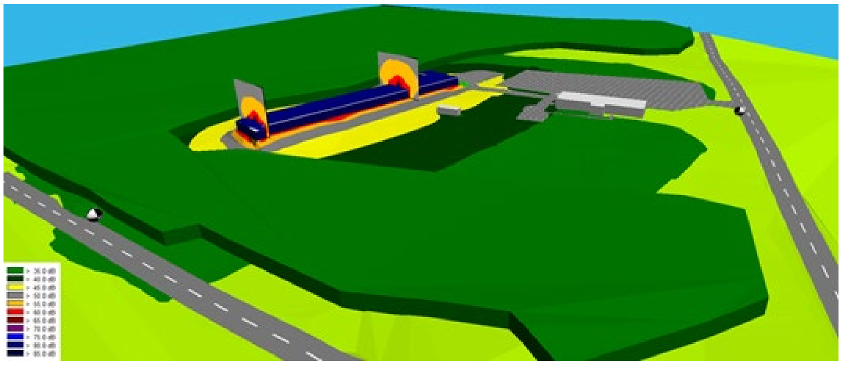
Sing Sing Correctional Facility Firing Range, Ossining NY LiRo designed the new Sing Sing Correctional Facility 8-lane Fully Enclosed Firing Range. This facility can accommodate up to 100-yard AR-15 qualifying in a fully enclosed environment. The site is located in a tight space, constrained by bedrock outcrops, a 70-foot high cliff, and the perimeter road and security wall. LiRo overcame the space constraints to meet all range, Preparation Suite classroom, access, utilities, and parking requirements by using the terrain as an advantage - stacking the classroom and covered parking below the Ready Area end of the range. The facility is equipped with state-of-the art ventilation and acoustical controls. The facility is equipped with an interlocking range control system and data is provided linking the security and mechanical systems to the main correctional facility information system. LiRo is presently performing construction services with construction underway and scheduled to be completed in the Fall of 2019.



Correctional Officer Training Academy Firing Range and Training Building, Coxsackie NY. LiRo has substantially completed design of an outdoor firing range and training building at the Coxsackie Correctional facility. The facility consists of a 5,000 square foot classroom training building for 60 trainees, a shooter's shelter, a steel escalator bullet trap, a 16-station 100-yard baffled firing range with shooting positions at 100-, 75-, 50-, 25-, 15-, 7- and 3-yards for AR-15 rifle, 9mm pistol, and shotgun training. The range is fully baffled (no blue sky) from 25-yards to target. The facility design also includes a separate CS-gas training building, and the main training building includes a heated outdoor access corridor with 6-eyewash stations to support 4-season capsicum spray training. Wick's Law compliant public bid documents for Construction, HVAC, Plumbing, and Electrical-contracts are in preparation with awards scheduled for late 2019.



Auburn Correctional Facility, Town of Sennett, NY. LiRo has completed the design and of an 8-lane 100-yard Indoor Firing Range for the Auburn CF. LiRo's work on the project began with development of program documents and permitting that included wetlands delineation and a SEQR long form, continuing though development of plans and specifications, preparing a SWPPP/NOI, and bid support. In response to resident concerns regarding range noise impact, LiRo collected background sound data, terrain, and ground cover information and modelled sound propagation from the range using the CADNA 3-D analytical model. The composite low bids for the project came in within 1-percent of the project budget estimate of $7.3 Million.

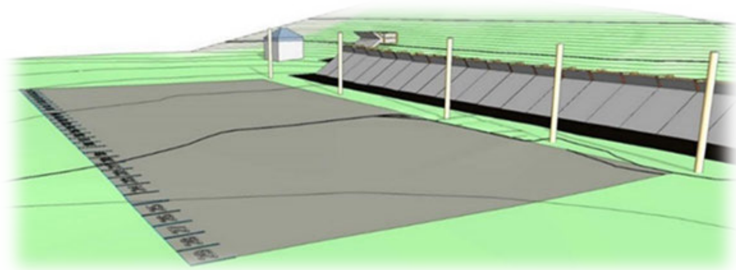


CADNA Model Sound Model Output - New Auburn CF Firing Range

Construction is underway and is on schedule to be completed in the Spring 2020. LiRo is providing ongoing construction services including SWPPP inspections, submittal review, RFI responses, Informational Bulletins, engineering inspections, and meeting support.

NYCDEP Beerston - Police Precinct Firing Range, Walton NY

Based on the New York City Department of Environmental Protection (NYCDEP) Stewardship Plan for this site, LiRo prepared a *Remedial Alternatives Analysis* in 2015 that evaluated alternatives for lead recovery and recycling from an earthen berm backstop. The alternatives combined an initial lead recovery event with varying combinations of technologies to reconstitute the range in a manner that enabled efficient and effective lead recovery and recycling events going forward. The five options included variations of earthen berms, a combination of berms and traps, steel escalator backstops, and granular rubber backstops. The preferred alternative, developed in close coordination with the police precinct to meet their qualifying requirements and the budget for the project , included a 29-lane small arms range backstop and a 300-yard .50 caliber rifle range backstop. LiRo then prepared *Plans and Specifications* in 2016 for the reclamation and backstop redevelopment and the project was bid publicly. LiRo *Construction Services* included submittal review, RFI responses, Informational Bulletins, and SWPPP inspections. Construction of the new range was completed in 2018.



Alternatives Analysis - Rendering of Selected Alternative

Beerston Police Precinct – Spring 2019. New 29 Lane Small Arms Range Bullet Trap – .50 Cal Trap lies to the East (right)

Dynamic Walk Down Range, Amherst NY

LiRo is designing a six-lane 65-yard indoor tactical pistol and rifle range within the Amherst Police Department's multi-function law enforcement facility renovation. The facility, located in an existing warehouse, will include a dynamic walk down range with control room/preparation area. The range is developed in an envelope that provides full ballistic containment and acoustic dampening. LiRo developed a CADNA 3-D analytical model for the range to access sound impacts to nearby residents including the range roof mounted HVAC equipment. The project also includes evaluation of additional space for development for 270-degree live fire training. The project is in the schematic design phase.

NYPD Coney Island Firing Range, Brooklyn NY

New York City Police Department (NYPD) Coney Island Firing Range Reconstruction, Brooklyn, NY. The range was originally built in 1996 for the New York City Transit Police and has been taken over by the NYPD. It consists of four shooting bays of 10 lanes on each side of a common control room. LiRo is providing engineering and design services for reconstruction of the NYPD's Coney Island firing range. LiRo is preparing and furnishing required contract specifications and drawings. The scope included investigations, inspections, documentation and certification, filing and permitting needed to prepare complete and accurate bid documents for the installation of new firing range systems. Estimated construction cost $5.5M. Bidding is anticipated to occur in 2019.

Noise reduction is a key issue. The major design feature will reduce noise in the space according to OSHA guidelines to obtain a reverberation time of less than 1:25 sec.

The site investigation included bullet traps, fire line booths, target retrieval systems, the tower range observation, communication and monitoring systems, and HVAC systems. A new modern, fully functional and safe firing range will be constructed in 2020.

References:

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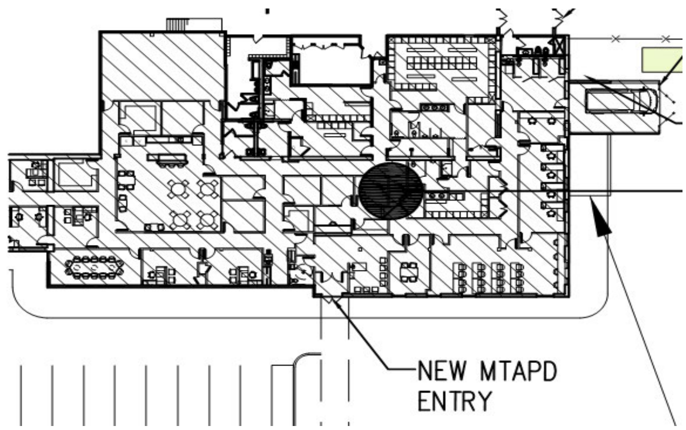
Additional Relevant Law Enforcement Experience

MTA Police Bethpage Facility



LiRo provided full MEP design services for a new police station in Bethpage, NY. The MTA police facility occupied 15,000 SF of an existing warehouse and office space facility. The first floor space required interior renovations, and upgrades to the existing mechanical and electrical systems. Along with designing to the latest NYS Codes, the design incorporates all requirements of the MTA Design Standards.

The mechanical design features three (3) packaged rooftop units, totaling 36 tons, along with supplemental DX split units and computer room air conditioning units for critical spaces. The rooftop units contained dehumidification capabilities, and separate steam humidifiers dedicated to each unit to ensure precise control of the interior environment. A variable volume and temperature (VVT) system installed is the ductwork from each rooftop unit allowed for individual temperature control of each zone within the facility. A building management controls system was designed to allow for remote control and monitoring of all equipment.

LiRo's electrical design included and upgrade to the existing service panels dedicated to the renovated space, as well as the installation of an emergency generator. New main distribution panels at 208V and 480V replaced existing panels, and a 48V/208V transformer and additional panels were added to serve all equipment and lighting. A 200 kw diesel emergency generator was designed to provide for full power backup of the police facility along with three new automatic transfer switches.

Nassau County Department of Public Works Prototype Design: 1st, 4th, & 8th Police Precincts

Performing design services for NCDPW under a term contract, LiRo provided a condition assessment survey of all eight county precincts, developed programming and prepared a schematic design of a prototypical police precinct that was used as a model for the redevelopment of all precincts.

Located at 900 Merrick Road in Baldwin, New York, the First Police Precinct was the prototype of Nassau County's "next generation" of police precinct buildings. It involved the acquisition of an adjacent property to allow for the expansion of the First Precinct's compound including phased construction of the new building, alongside the operational precinct building, followed by the expansion and improvement of the police compound. The schematic design included the development of a prototypical Police Precinct based on the Police Department's approval of the program report. Diagrammatic studies showing spatial relationships of rooms and areas to services and utilities were prepared. The design incorporated flexible space planning for future expansion needs. An estimate of the probable cost of the project was prepared and computerized Building Information Management (BIM) was developed. This 25,000 square feet facility was completed in 2016.

Located at 1699 Broadway in Hewlett, New York, the existing Fourth Police Precinct site is approximately 1.18 acres in size. The project consists of the demolition of the existing precinct, new ground-up construction of the Fourth Precinct and re-construction of the site's parking lot. The new station house will be a 25,000 sf structure consisting of 2 stories of 10,000 sf each designed to meet current and future police operations plus a roof penthouse of approximately 5,000 sf for mechanical equipment, telecommunications room and police record storage. The Fourth Police Precinct was completed in 2018.

Located at 286 N. Wantagh Avenue in Levittown, New York, the existing Eighth Precinct site is approximately 1.9 acres in size. The project consists of the demolition of the existing garage and existing station house at the Eighth Precinct site, the construction of a new station house and garage, and the re-construction of the site's parking lot.

The new station house will be a 30,600 sf structure consisting of 3 stories of 8,600 sf each designed to meet current and future police operations plus a 4th floor of approximately 4,800 sf for mechanical equipment, telecommunications room, fitness gym and police record storage. The Eighth Police Precinct was completed in 2017.

Niagara Falls State Park Police Precinct

This project involved the construction of a 9,000 SF new Police Station Facility for the Niagara Falls State Park Police. The project is a Wicks Law project with four separate prime contractors performing General Construction, Electrical, HVAC, and Plumbing work.

This one story building consists of office and administration spaces, locker and fitness rooms, an interview room, booking and evidence areas, radio room, bathrooms, and a kitchen. The structural composite of the building is masonry bearing structure with steel truss hip/dormer roof and clear story atrium. Finishes included exterior stone veneer and lap siding, asphalt architectural roof shingles, gypsum wallboard system partitions, and acoustical ceilings. A new asphalt parking lot for officer, civilian employees, and visitor parking. New utilities include, electrical, telephone, storm, sewer, and water services. The site will be finished with new landscaping and sod.

LiRo provided preconstruction, construction, and post construction management services for this project. During the preconstruction phase, we performed 3 estimates. One during the design development phase and two estimates during the construction document phase of design. Additionally, LiRo performed constructability review during the preconstruction phase. As well as inspection services and project administration during the construction phase.

Recent City of Buffalo Experience

City of Buffalo City-Wide Inspection Services for Various Projects

LiRo was selected by the City of Buffalo, Department of Public Works, Parks & Streets to provide highway engineering services for an array of projects for the period of 2015 to 2018. Services provided included Detailed Design and Bid Packages, construction inspection and management and construction cost estimating for various rehabilitation, reconstruction project candidates for Public Work Budgetary development. LiRo's staff also provided Engineering support services to the City Engineer in pavement evaluations along with providing alternative options with construction cost estimate for brick pavement rehabilitation and preservation.

Western New York Workforce Training Center

LiRo was retained by Buffalo Urban Development Corporation (BUDC) to complete a Remedial Investigation (RI) and Alternatives Analysis (AA). The investigation was conducted under the guidelines of the New York State Department of Environmental Conservation's (NYSDEC) Brownfield Program at the Western New York Workforce Training Center, located at 683 Northland Avenue in Buffalo, New York. The property is currently owned by 683 Northland, LLC. The RIAA was performed under a Brownfield Cleanup Agreement (BCA) between 683 Northland, LLC and the NYSDEC. The BCA required 683 Northland, LLC to investigate and remediate potential areas of environmental concern associated with the site. LiRo also prepared a Remedial Action Work Plan (RAWP) for site remediation.

The site is approximately 8 acres with approximately 235,000 square feet (sf) of buildings that were constructed between 1911 and 1983. The building complex is comprised of a four-story office area on the north side along Northland Avenue, a series of ten connecting manufacturing spaces, and a detached one-story shed located on the west side of the facility.

The alternatives assessment (AA) summarized the findings of the RI: The nature and extent of contamination and the current and future exposure pathways at the site; identified statutory or regulatory remedial action goals (both Federal and State); identified remedial action objectives (NYSDEC Part 375 Commercial Soil Cleanup Objectives); screened no action; containment; in-situ and ex-situ remedial technologies.

LiRo assembled the technologies into remedial alternatives and evaluated the alternatives using NYSDEC's two threshold criteria and six evaluation criteria. LiRo then identified the selected remedy as excavation with off-site disposal of soil with PCB contamination greater than 10 parts per million and a cap cover system over impacted soil remaining after implementation of the two IRMs. LiRo then implemented a soil vapor barrier in the decontaminated facility, monitored natural attenuation for groundwater and a Site Management Plan that includes property use restrictions, soil management and Site monitoring.

The AA was summarized in a RI/AA report that was prepared under a LiRo NYS licensed Professional Engineer. The AA followed the requirements of NYSDEC DER-10 Technical Guidance.

Consultant Fee Schedule 2

LiRo's proposed fee is presented below. The fee includes all subcontractor and direct costs and is based on complete design of the 12,000 SF proposed facility on the identified property and all of the scope of work elements for each phase described by the RFP.

Request for Proposals
BUFFALO POLICE SHOOTING RANGE FACTILITY
JOB # 1917

		LUMP SUM FEES		ANTICIPATED HOURS	
1.	Pre-Design, Scoping & Programming Phase	$ 4,080	LS	30	Hrs
2.	Site Development Phase	$ 7,460	LS	104	Hrs
3.	Schematic Design Phase	$ 32,612	LS	350	Hrs
4.	Environmental Phase	$ 8,000	Hourly N.T.E.		
5.	Design Development Phase	$ 38,064	LS	372	Hrs
6.	Construction Documents Phase	$ 54,264	LS	539	Hrs
7.	Bid Phase	$ 6,412	LS	46	Hrs
8.	Construction Administration/Closeout	$ 45,764	LS	488	Hrs
	FEE SUBTOTAL / HOURS SUBTOTAL	$ 196,656	LS	1929	Hrs

TOTAL CONTRACT FEE $ 196,656 **LS**

NOTES

1. The above fee schedule shall be fully completed and included with your proposal. Payments shall be made, at minimum, monthly based on the percentage each task has been completed. In the event a task is not required or the project is canceled, the Consultant shall be paid only for work completed.

2. The scope of work listed in this RFP supersedes all submitted proposals, unless the RFP has been modified by addendum or written documents from the owner.

3. Fees quoted for each item shall **include** all costs associated with the item, including, but not limited to: mileage, reproduction, printing, phone calls, and postage.

4. Prices quoted shall be based on the described Scope of Services and an estimated total construction plus A/E costs of **$ 1,100,000.00**

Availability 3

LiRo is available to begin working on the project immediately upon authorization to proceed from the City of Buffalo.

LiRo is fully capable of meeting the response time requirements and, in fact, we have distinguished ourselves by providing extremely timely assignment responses. LiRo has the depth of staffing and equipment, locally, from our Buffalo office, to address the requirements of this project. This project will be performed and managed from our staff of 70 employees located in our downtown Buffalo Office.

The assembled project team is experienced in all professional aspects needed for the project. This includes our local staff as well as two additional Buffalo based subcontracted firms, Foit-Albert Associates and Frandina Engineering and Land Surveying, PC, and two local subcontracted firms Trophy Point Construction Services and Consulting (Blasdell) and Nature's Way Environmental (Alden). Our local in-house firing range designers, environmental professions, mechanical, electrical, and plumbing engineers, civil and transportations engineers is the same team that managed and/or performed the projects presented in the previous experience section of this proposal.

The LiRo Team has a pool of additional resources to draw from to ensure not only that all the requirements of the project are met, but that the best and most appropriate personnel and equipment are provided to the City of Buffalo.

The LiRo Project Team and LiRo management understand the financial constraints associated with capital work and the need to complete work on time, cost effectively, within budget. It is with enthusiasm, great interest, and above all our total corporate commitment to the City, DPW, and BPD that we propose to serve as your design consultant for the Seneca Street project. LiRo stands ready to start immediately on this critical assignment and we commit to safely deliver a project that is of high quality, on schedule and within budget.

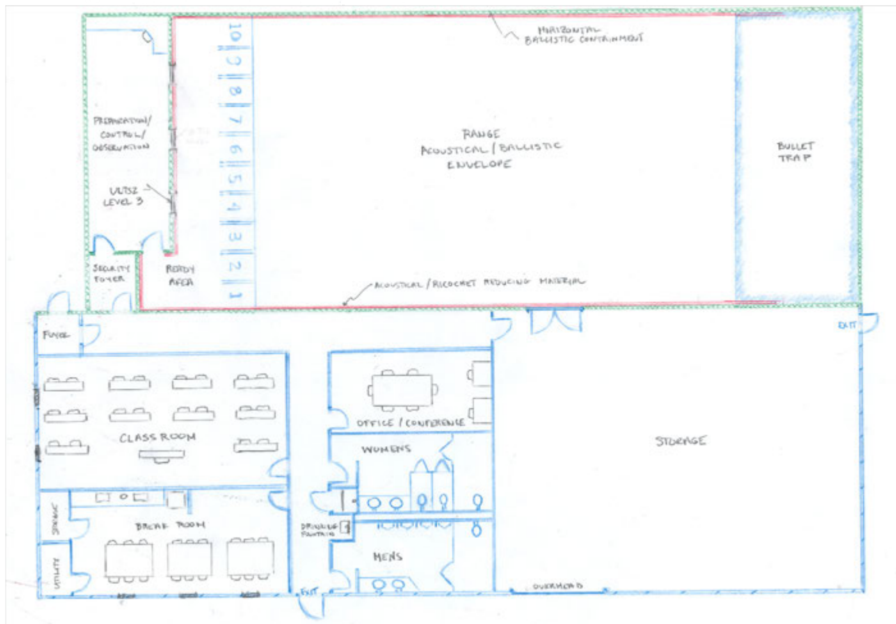
Understanding of the Scope of Work and Approach 4

LiRo will develop concepts and design for the planned indoor Buffalo Police Shooting Range Facility that meet the following design objectives:

- Provide a facility that includes a shooting range, classroom space for 20 occupants, office space, break room, restrooms, mechanical support rooms, fire protection, and up to 5,000 square feet of storage space
- The building shell will be a pre-engineered metal structure accommodating 12,000 square feet of space.
- Provide space for live fire handgun training from a fixed firing line including a target retrieval system to vary target distance and to enable target change-out from the fixed firing line.
- Provide for a clear span for unobstructed firing from 25-yards to target for a 10-firing lane wide range
- Provide a geometric configuration supplemented with ricochet reducing materials to create a space safe for trainees and instructors
- Fully contain fired projectiles within the range to assure protection of the community and spaces in the facility outside of the range
- Create a noise and reverberation reducing environment that provides comfort to users within the range and within the facility
- Minimize sound propagation from muzzle noise within the range and noise created by HVAC equipment to levels below acceptable municipal noise levels
- Design a ventilation system that effectively removes gun smoke away from the shooters to minimize lead exposure and includes an easy maintenance filtration system that captures lead before discharge outside
- Design a central range control system that integrates and sequences the lighting, ventilation, targeting controls, audible/visual alarms, and access controls
- Provide lighting to create a safe and effective training environment
- Include lighting, ventilation, and bullet containment systems that are easy to maintain for which maintenance costs are minimized
- Provide sufficient access and egress to enable maintenance and recycling
- Provide security access controls to the facility and to the range
- Create an operating space and systems that comply with applicable OSHA, National Electric Code, NYS Energy Conservation Code, International Building Code, and NYS Fire Safety codes
- Provide a design with a projected construction cost that falls within the City of Buffalo budget for the project
- Provide building site development in the context of the City's parking requirements
- Address permitting requirements including provision of SEQR documents and development of a Storm Water Pollution Prevention Plan (SWPPP)

Range Configuration - LiRo will base the range configuration and fit out on the training requirements and goals of the Buffalo Police Department. LiRo will meet with BPD representatives to clearly discern those goals and requirements and will design draft concepts for further review. A preliminary opinion of construction cost will be generated for each selected concept and the pros and cons of each concept will be presented to the Department. Each concept will meet the objectives outlined above.

The function of and requirements for the range will be carefully reviewed with the BPD not only to determine current firearm use, distance, targeting, lighting, training requirements, range controls, data, and communication, but also to anticipate possible changes in firearms use and revision of training techniques. Ammunition, target, spent brass storage requirements and shot volume will be similarly evaluated given current requirements and potential future changes. Additionally, access and security requirements will be established and coordination with the BPD. Spatial and interactive relationships between the range and the other facility functions, and the use requirements of the large storage area, will be coordinated with the DPW and BPD. A potential layout for the required space is presented below.



Ballistic Containment - Fired bullets will be fully contained within the range in a concrete and steel ballistic enclosed envelope (ceiling-sides-floor) that directs bullets to a bullet trap. The concrete floor and walls (minimum of 8-inches solid concrete or concrete filled CMU) will provide bottom and lateral containment. Vertical containment will be accomplished by hanging AR-500 steel plates from steel joists. The plates will be angled to deflect stray bullets downrange and to provide protected openings for lighting and ventilation. Joists will be set below the roof joists of the pre-engineered building shell. Additional bullet containing and noise reducing materials may be added on top of the joists to provide a continuous ballistic and sound envelope.

Bullet containment within range will need to be capable of handling a large volume of fired rounds before recycling is required. The two general types of available traps are a steel scroll trap and a granular rubber trap. The advantage of a steel scroll traps is that there is no need to occasionally remove rubber particles from the target area, no need to replenish the granulated rubber, and the projectiles are conveniently collected in buckets behind the trap. However, steel traps are noisier and generate lead dust. A target dust collection system is mandatory with this design to capture the dust otherwise it would be captured by and overload the range ventilation system requiring more frequent filter changes a great expense. A maintenance corridor would be necessary behind the bullet trap. This type of trap has a larger footprint than a rubber trap. A granular rubber trap generates little to no lead dust. The rounds can be extracted virtually intact. This type of trap can hold about 70,000 124gn round/lane (1,240 lbs.) of lead per shooting position before recycling is required. This type of trap is less noisy than a steel trap. The target area would need to be swept occasionally to remove rubber particles. Granular rubber traps are considerably less expensive than steel escalator traps (about 25% of the cost). Because of the smaller footprint and less cost, it is likely that a granular rubber trap is the best application for the project.

There are several bullet trap manufactures that supply at least some of the traps identified above. Each of the manufacturer's traps have different advantages, disadvantages, space requirements and associated costs. Once the specific requirements are known, LiRo will contact the manufacturers to develop site specific selections and provide estimated costs for DPW and BPD review. A partial list of manufactures includes Action Target (Provo UT), Savage Range Systems (Westfield MA), Meggitt Training Systems (Suwanee, GA), and Range Systems (New Hope, MN). LiRo has experience working with all the listed manufacturers.

Targeting Retrieval Systems – Target retrieval systems can simple or complex. At minimum, the system must be armored and durable and meet the design criteria to vary target distance and to enable target change-out from the fixed firing line. More complex systems can incorporate features such as a touch screen interface, programmable variance of target lighting, target exposure, and distance, and strobe lighting. The systems evaluated will meet the training requirements of the BPD in the context of the project budget.

Shooting Stalls - Shooting stalls and benches are available form a variety of manufactures. Various shooting stalls will be included in the preliminary range concepts.

Integrated Controls - HVAC, lighting, and targeting systems will be integrated to provide clear signals that the range is safe to use. For example, lighting and targeting systems will not be enabled in firing ready state until the HVAC system is on for a sufficient time such that air movement can remove gun smoke away from the shooters. In the ready state, targeting systems and lighting variations can be controlled by a range manager in the control room or by an in-range instructor operated with a hand held controller. The ability for a range manager to observe range activities visually can be incorporated into the design by adding ballistic safe windows up range in the control room.

Communications - Communications between range control, instructors, and shooters and procedures for ingress security are important components of range design. Once the specific BPD communication requirements are known, communications options will be costed and included in the preliminary concepts for DPW and BPD review.

Containing Lead Within the Range - To be protective of lead exposure, LiRo's design will incorporate features, when combined with management practices, create a space and system that is protective of the environment, the community, shooters, and maintenance personnel. Lead fumes, dust and gun smoke from firearm muzzles will be contained within the range by a dedicated ventilation system (not used to service other areas of the building) that maintains a negative air pressure relative to the outside environment and to the other sections of the building. Pressure is continuously monitored to verify proper operation. The range ventilation system will recirculate a portion of air and exhaust the rest in order to maintain the negative pressure environment. Air will be filtered prior to exhaust/recirculation with a triple filter system that effectively removes all (99.97%) of lead particulates. An interlock system (lighting, targeting, & ventilation) will prevents the range from being used without the ventilation system active. The filters will require routine replacement. The filter pressure differential is cautiously monitored and will trigger a maintenance warning when the filters need to be changed.

Acoustical Design - The sound from a firing range is a potential nuisance to the community and a source of discomfort to the shooters and instructors and must addressed by the design. The range buildout will be designed to efficiently provide a combination of conventional and state-of-the-art sound reverberation and ricochet reducing materials such that the breakout noise due to the firing of weapons will be 55-60 decibels or less at the outside surface of the exterior walls. The noise generated by the dedicated outdoor air ventilation system will also be considered.

Architecture - As currently envisioned, acoustical and ballistic containment will be achieved by building a CMU block wall range structure within a pre-engineered steel building shell. Other functional areas will be integrated efficiently within the steel structure. The facility must safely meet all functional requirements of the DPW and BPD including ingress security, be durable, and the layout and appearance of the building must meet Planning Board requirements. The facility should be ergonomic and comfortable in the context of the project budget.

Public Art – Our team will coordinate with the Buffalo Arts Commission regarding recommendations on incorporating public art as a component of the view shed of the facility.

Project Execution - The key to a successful project is establishing a clear and detailed understanding of scope very early in the project. Our philosophy is to maintain the same team on a project from its inception to completion. This provides continuity and streamlines the project in house. LiRo specializes in the design of critical spaces with tight budgets and pressing schedules. As a matter of project course all discipline leads meet weekly to coordinate their design - at LiRo we have found coordination between trades is best done as the design develops.

Project Specific Approach - We have tailored a project specific approach to the list of tasks and requirements outlined in the RFP. Using each of the tasks identified, we have assigned staff hours to the project and have generated a fee for the project including hours and fees for our subcontractors.

Pre-Design, Scoping and Programming Phase - The project scope is to design a facility that includes a shooting range, classroom space for 20 occupants, office space, break room, restrooms, mechanical support rooms, fire protection, and up to 5,000 square feet of storage space. LiRo will schedule a kick off meeting with key stakeholders from the City of Buffalo. We will start by reconfirming the needs of the project, the goals, the technical challenges, and the physical parameters of the building. Additionally, the BPD has personnel with expertise in operating and training within a shooting range environment. We recognize that this knowledge is extremely valuable and will provide us with a solid foundation. Prior to the meeting, the use of pre-fabricated shooting lanes will be evaluated and the results presented at the meeting. Our team, including representatives of both LiRo and Foit-Albert, will visit the site and confirm conditions. Also, a line of communication will be opened, in coordination with the DPW, with City Planning staff and with Arts Commission staff to discern their expectations for the project. Upon approval of the scope of work, we will develop a milestone schedule and MOU for the project for submission to the City of Buffalo.

Site Development Phase - LiRo will coordinate with our subcontractors to conduct a topographic survey and to obtain geotechnical data needed to adequately complete the civil and structural design of the facility. Any additional destructive testing will be coordinated with the DPW.

Schematic Design - LiRo and our subcontracted architect (Foit-Albert Associates) will develop schematic design elevations, plans, sketches, and memoranda that will convey design concepts to the DPW and BPD. During schematic design development, the LiRo team will maintain an open line of communication with the DPW to review samples and cut sheets and relative costs so that key materials and equipment used in the concepts presented are pre-vetted to streamline DPW and BPD review of the schematic design. Meetings will be scheduled with the DPW as needed. LiRo will record the minutes for all meetings. The SEQR process will be started in this phase so that any potential impediments or needs for additional information are identified and address before proceeding to the later stages of design.

The preliminary itemized cost estimate, developed in this phase, will determine if the project is proceeding as a single contract or multiple prime contracts, based on overall contract values. This will be confirmed with the City. We will schedule a meeting with the DPW for a preliminary review of the schematic design, which will be accompanied by a Code Review Checklist or letter report. This will allow the Permit Office to comfortably familiarize themselves with the project and the design team to address any code issues that may arise from this preliminary review. Alternates will be identified for project budget management. A potential redesign for a scaled down facility at the Design Development Phase is budgeted in our proposal. This redesign will be contemplated by the Schematic Design.

Environmental Phase - LiRo will implement the recommendations of the Phase II Environmental Assessment (not yet available). Pricing for the environmental phase will be negotiated with the DPW based on the unit rates presented in this proposal.

Design Development Phase - Drawings will be prepared in detail during this phase. All work will be reviewed by Trophy Point, for development of a Design Development phase Third Party Certified Cost Estimate. The team will visit the site, in coordination with the DPW, as needed to advance the design.
Once the drawings have advanced in sufficient detail for a permit, we will submit them to the City of Buffalo to begin the review process for permitting. The cost of permit application is included in our fee as part of a $250 allowance against which the permit filing fees can be drawn. Final permits are released to licensed contractors in the City of Buffalo, so we have included the filing fees of $250, which we are including in our proposal.

Official submission will be made to the Planning Board during this phase to garner approval. This submission will also include the public art component. SEQR documents will also be readied for submission and a draft SWPPP will be prepared. Steps for approval by SHPO, the Zoning Board, Historic Preservation Board will be advanced in this phase.

The design and estimate will be reviewed with the DPW team to determine conformance with the project budget, and modifications to the scope will be incorporated as appropriate prior to the advancement to Construction Documents. Meetings will be scheduled as necessary with the DPW. LiRo and Foit-Albert will also meet with Buffalo's sustainability personnel to discuss possible energy incentives.

Construction Documents - Upon authorization to advance to the Construction Documents phase, the design team will complete the construction documents for competitive bidding. This will include all plans and specification sections for each project and phase required for bidding. Up two three separate bid packages may be required and alternates will be included to phase for budging purposes.
We will meet with the City of Buffalo Project Manager for review of all front-end requirements for the City, as well as development of the draft contract for inclusion in the project manual. This will include the first draft of the project advertisement (for finalization of bid dates once the project is ready to be let), the bid forms and the project specific operations requirements. It is our goal to have all of the scheduling, phasing, operational and contractual issues clearly defined as part of the project specification by the end of this phase

LiRo will provide a Statement of Special Inspections and final comments from the City of Buffalo Plan Review will be incorporated to clear the project for final issuance of a building permit to the awarded contractor. Design coordination meetings will be held with the DPW and BPD and the site will be visited as necessary in this phase. Color preconstruction photographs will be taken and cataloged during one of the site visits.

The Third Party Certified Cost Estimates, separated by package and alternate, will be updated at 95% document completion. These estimates will be submitted after review to the City for final approval in advance of release of documents for bidding.

LiRo will obtain approvals from SHPO, the Zoning Board, Planning Board, Historic Preservation Board, Buffalo Plan Review, and relevant public utilities before the bid phase.

Bid Phase - A Bid Advertisement will be developed in conjunction with the City of Buffalo Project Manager (which will be included in the project manual) and will include all necessary bid dates, times and pre-bid meetings. This will be provided to the City of Buffalo for insertion in the Buffalo News or other publication. Up to 20 copies of the bid documents will be made available at a nonrefundable fee (paid to the City) of $58. Four additional copies of the specifications and addenda and one set of drawings will be provided to the City. Electronic documents will be provided on jump drive. We will maintain a distribution log of all prospective bidders to document receipt and issuance of addenda.

We will hold a pre-bid walkthrough on site with the City and potential bidders, to discuss project with those bidders, issuing minutes from that meeting in our first of the required addenda. Additional addenda will be issued to address any contractor questions or concerns. All addenda will be posted in the specification book for development of the project contract through Corporation Counsel's office.

We will attend the bid opening, providing bid tabulation forms for bidder use, as well as recording all bids for review and comparison. We will interview apparent low bidders and make recommendation to the City of Buffalo of the lowest responsible bidder.

LiRo will make revisions to the contract documents as may be required as a result of bidding cost overrun for rebidding. Addenda will be incorporated into the revised documents.

This phase also includes a post-bid meeting with all prime contractors 1 week following bid opening and prior to contract execution.

Construction Administration - LiRo is one of the leading construction management firms in New York State and we will leverage our experience in the context of the City's and the project's particular requirements to successfully administer the construction work to successful completion.

Using the photographs we have taken throughout the design document development, and supplementing it with additional scope photographs as may be required, we will develop a submission of pre-construction photographs keyed to plans. Post construction photos from the same location will be cross-referenced in the same report.

LiRo will coordinate a pre-construction meeting and conduct bi-weekly progress meetings throughout the period of construction. Procedures for submittals, shop drawings, invoice review, Requests for Information (RFIs), modifications, schedule tracking, and testing will be reviewed at the kick-off and then reviewed at each bi-weekly meeting. LiRo will prepare minutes for each meeting for distribution.

We will review shop drawings providing clarifying and coordinating details, and will review submittals, keeping an organized log of shop drawings and submittals. We will review and answer all RFI's and maintain RFI log. As necessary, certifications of contractor personnel will be verified.

An experienced inspector, appropriate to the construction activity ongoing at that time, will visit the site weekly to assure compliance with the contract documents and design concept. LiRo will provide 3rd party testing, inspections, reporting, and certification per the approved statement of Special Inspections and per Chapter 17 of the NYS Building Code. We will identify and address unanticipated field conditions and issues promptly and take appropriate action and field directives based on observations and test results.

We will review and leek a log of contractor invoices, provide the contractor with a pencil copy markups, and submit the revised invoices to the City for payment. We will notify the DPW of any potential change orders or schedule impacts and evaluate the justification and reasonableness of any change orders requested by the contractor.

Throughout the construction process, our Project Manager will communicate frequently with the DPW regarding all relevant aspects of the project.

Contract Closeout - Upon construction completion, LiRo will secure a general release, a maintenance bond, required insurance certifications, and closeout documents from the Contractor. The general release must include downstream releases from the contractor's subcontractors. LiRo will also obtain O&M data, diagrams, replacement parts lists, guarantees, and manufacturers service directories to the DPW in one hardcopy volume and electronically.

Prior to final contractor payment, we will make a final inspection of the site to assure that it is in first class working condition and certify to the City that the project was completed in accordance with the plans and specifications, as clarified by RFI responses, approved modifications made during construction.

We will provide an organized and indexed hard copy of record drawings and all other records, such as field notes submittals, RFI's, and photographs to the DPW for their record. This complication will also be provided electronically.

Quality Control - The LiRo team will assure that the requirements of all applicable regulations, Contract requirements, and DPW's requirements are met. We are ISO 9001/14001 compliant. Our QA/QC will be led by our QA/QC Officer, Mr. Frank Ulisse, PE, and will achieve project documentation and record keeping in accordance with the City's, as well as our own QC Plan. Team members have been audited numerous times by public clients and regulatory agencies and understand their commitment to quality. Our internal project specific QA Plan will be developed at project initiation with the understanding that the DPW and City dictate the use of specific procedures, and that project organization titles and responsibilities vary amongst our projects. Our QA/QC Officer will work with our Project Manager to ensure that:

- All aspects of the Team's contractual obligations are being fulfilled in a manner consistent with your requirements.
- The standards established in the Team's QC Plan are maintained within the framework of the project.
- Project personnel understand their responsibilities and are equipped to fulfill them.

Resumes of Key Personnel 5

The organization of our project personnel is presented below. Resumes of key LiRo and subcontractor personnel are also included in this section.

Principal-In-Charge
Robert Kreuzer, PG

QA/QC
Frank Ulisse, PE, PMP

Project Manager
Michael Lydle, PE

Key Personnel

Foit-Albert Associates (MBE)

Architecture
Gwen Howard, RA
Shawn Cowie, RA
Scott Archambault, RA

Structural Engineer
Jason Ciurzynski, PE

Firing Range Design Experts
Bruce Przybyl, PG
Richard Blanch, PE

Survey/Geotechnical
Frandina Engineering and Land Surveying, P.C. (WBE)
Rosanne Frandina, PE, LS
Mark Childs

Environmental
Steve Frank, PG
Natures Way Environmental

Mechanical/Electrical
Chris Abramo, PE, LEED AP BD+C
Robert Eimer, PE
Joseph Hallmark, PE
Daniel Cimino, CPD
Pavel Belavus

Cost Estimating
Trophy Point Construction Services and Consulting (SDVOB)
Joe Dommer

Robert Kreuzer

Principal-in-Charge – WNY Operations Manager

Education

B.S., Geological Science, State University of New York College at Buffalo

Licenses

Professional Geologist, New York

Certifications

OSHA 40 Hazwoper Certified

OSHA 30 Certified

OSHA 10 Certified

OSHA Confined Space Entry Certified

Underground Storage Tanks

PROFESSIONAL PROFILE

Mr. Kreuzer is a LiRo Senior Vice President who has over 29 years of experience in site evaluations, corridor studies, environmental sampling, soil management, supervision of field activities, aquifer testing, groundwater modeling, project coordination, and report development at sites throughout New York State as well as hazardous waste sites, industrial sites, and commercial sites. His experience also includes asbestos, lead, decommissioning / demolition projects, petroleum remediation projects, and industrial waste management projects. He has worked under the New York State Department of Environmental Conservation (NYSDEC) Superfund and at Voluntary Cleanup Program (VCP), Brownfields, Environmental Restoration Program (ERP) and petroleum consent order sites. During his career, he has demonstrated an exceptional ability to manage and coordinate a wide range of environmental projects for public and private sector clients throughout the country, with the majority of the work being located in New York. He has been involved with numerous projects involving Federal agency remedial efforts (i.e., United States Environmental Protection Agency [USEPA], United States Army Corps. Of Engineers [USACE], Air Force Center for Engineering and the Environment [AFCEE]) and NYSDEC site characterization and remediation programs.

EXPERIENCE

Sing Sing Correctional Facility, Outdoor and Indoor Firing Ranges. Served as Contract Executive for multiple phases of evaluation and planning for NYSOGS and DOCCS starting with evaluation of facility's existing outdoor range and finishing with design of a the new 8-lane Fully Enclosed Firing Range. This replacement facility can accommodate up to 100-yard AR-15 qualifying in a fully enclosed environment. The evaluation of the existing range concluded that most of the range components had reached their serviceable lifespan and LiRo recommended removal and replacement. The new indoor range, is located in a tight space, constrained by bedrock outcrops, a 70-foot high cliff, and the perimeter road and security wall. LiRo overcame the space constraints to meet all range, Preparation Suite classroom, access, utilities, and parking requirements by using the terrain as an advantage - stacking the classroom and covered parking below the Ready Area end of the range. The facility is equipped with state-of-the art ventilation and acoustical controls. Construction is underway..

Correctional Officer Training Academy, Range and Training Building, Coxsackie, New York. Contract Executive for the preparation of a Program Report for NYSOGS and DOCCS including code review for a new firearms training facility at the Coxsackie CF. Permitting included wetlands evaluation, archeological permitting, and SEQR Forms. Managed design and coordination of bid document preparation of an outdoor firing range and training building at the site. The facility consists of a 5,000 square foot class room training building, a shooter's shelter, a steel escalator bullet trap, a 16-station 100-yard baffled firing range with shooting positions at 100, 75-, 50-, 25-, 15-, 7- and 3-yards for AR-15 rifle, 9mm pistol, and shotgun. The range is fully baffled (no blue sky) from 25-yards to target. The facility also includes a separate CS-gas training building, and the main training building includes a heated outdoor access corridor with 6-eywash stations to support 4-season capsicum spray training. Bid documents are in preparation and construction startup is scheduled for 2020.

Auburn Correctional Facility - Town of Sennett, New York. Contract Executive for design of an 8-lane 100-yard Indoor Firing Range for the Auburn CF. In response to resident concerns regarding range noise impact, LiRo collected background sound data, terrain, and ground cover information and modelled sound propagation from the range using the CADNA 3-D analytical model. The range design is based on the Sing Sing CF range template fit to the terrain, view shed, and facility requirements. Construction is underway.

Robert Kreuzer

Senior Vice President – WNY Operations Manager

Buffalo Urban Development Corporation – Northland Redevelopment, Buffalo, NY: Project Principal for the Northland Corridor project, which comprises 12 properties and 50 acres of a former industrial district within the East Delavan-Grider Brownfield Opportunity Area in Buffalo, NY. LiRo has been contracted by the Buffalo Urban Development Corporation to formulate a redevelopment plan for acquiring multiple properties that will eventually lead to returning the Northland Corridor to productive use, attract new businesses, assist with revitalizing the surrounding neighborhood, and provide employment opportunities for nearby residents by creating a new manufacturing hub on Buffalo's East Side. The project includes real estate/ market analysis, SEQR documentation, historical preservation, planning/ transportation/ infrastructure analysis, remediation cost estimating and community participation to culminate in a detailed Redevelopment Plan.

City of Buffalo 377 Main St. & 369 Washington Environmental Assessment and Cost Estimating for Demolition of 2 Buildings - LiRo provided professional engineering and environmental consulting services for the evaluation of structural and environmental conditions for the former AM&A's store and warehouse complex in downtown Buffalo. The project included detailed inspections and evaluations necessary for the development of cost estimates for the proposed abatement, remediation and demolition of the subject buildings. The project buildings consist of a 10-story former retail building of approximately 300,000 sf and a five-story warehouse building of approximately 110,000 sf.

The project involved focused inspections and evaluations of structural conditions of the building and environmental inspections to identify and quantify asbestos and hazardous materials. The highlights of this project are summarized below.

- An evaluation and field inspection to validate the location and quantities of previously performed asbestos survey
- Structural evaluations to determine the effects of the proposed demolition on adjacent building and structures
- An inspection and evaluation of hazardous materials located throughout the buildings (i.e. PCB light ballast/electrical equipment, freon, USTs, and deteriorated lead paint)
- Shoring evaluations for the protection of adjacent foundations
- Evaluations of the demolition approach and requirements

Detailed cost estimates for asbestos abatement, hazardous materials removal, foundation shoring and building demolition.

Empire State Development, Episcopal Church Home Abatement and Demolition, Buffalo, NY: Project Principal for the abatement, demolition, and site stabilization of a former geriatric complex on behalf of the Empire State Development Corporation. Managed the development of Plan and Specifications, prepared the Environmental Assessment form, participated in public forums, and prepared community new letters.

ECHDC Buffalo Memorial Auditorium and Donovan Office Building Construction Management Services, NY: Project Manager for construction management services of environmental projects for the Buffalo Memorial Auditorium (AUD) and Donovan Office Building (DOB) LiRo has been retained by the Erie Canal Harbor Development Corporation (ECHDC) to perform construction management services to coordinate and oversee asbestos abatement and environmental remediation work at the AUD and the DOB in downtown Buffalo, NY. The AUD is a 200,000+ S.F. arena that was constructed in 1939. The DOB is a 160,000 S.F. eight story office building. The AUD and DOB properties are the proposed site for a mixed-use retail complex including structured public parking facilities. The objective for this project is to perform all asbestos and hazardous materials abatement to allow for building demolition under a subsequent contract. The performance of the asbestos abatement and hazardous materials removal work under the proposed project schedule of 18 months is a critical element of the overall project schedule for demolition and site redevelopment.

Erie Canal Harbor Development Corporation, Canalside Project, Buffalo, NY: LiRo provided CM services for the initial development on the 5 acre site which was formally the site of the Memorial Auditorium. LiRo overcame multiple subsurface field challenges in working with local contractors in order to install deep foundation systems over this historical urban site. Complex filtration and water treatment systems were installed and routed from underground mechanical rooms at both sites. LiRo also worked hand in hand with the owners A/E team in order to ensure that onsite quality control tolerances were met on all facets of these unique construction projects. LiRo's CM team oversaw the installation and work of more than $2 million of custom fabricated granite canal walls and site retaining walls.

Former Trico Plant, NY - Managed the development of the Draft Environmental Impact Statement (DGEIS) for the adaptive reuse of a 2.1 acre, six story, multiple building former manufacturing plant located in a downtown urban location in Buffalo, New York. The facility is listed on the National Registry of Historical Places, is in a state of disrepair, and contained asbestos, lead paint, and other hazardous materials. The proposed plan included partial demolition of the historic structures and incorporation of remaining historical buildings into a new technology center. The project required completion of all aspects of the SEQRA process and submission of a DGEIS that incorporated multiple master plans. The project was completed on a very short schedule.

Erie Canal Harbor Development Corporation – Buffalo Outer Harbor Wilkeson Pointe Park Improvements: Project Principal for the redevelopment design of two parcels on Fuhrmann Boulevard located on the Outer Harbor in Buffalo, New York. The project consisted of preparing approximately 15 acres for future development along with providing amenities for public access. The work included shoreline enhancements, public walkways and bike paths, water taxi docking facilities, landscape features, parking, site lighting and the rehabilitation of an existing metal building for use as a comfort station and storage building. Landscape features included rain gardens and retention ponds for runoff. Previous site usage and contamination required placement of an earthen cap over the entire site and removal of an underground petroleum storage tank and the associated piping.

Michael F. Lydle, PE

Project Manager

Education

B.S., Civil Engineering, State University of New York at Buffalo

A.A.S., Civil Technology, Erie County Community College

Licenses/Registrations

Professional Engineer, New York

Certifications

OSHA 10 Hr. Certified

NYS DEC 4 Hour Erosion and Sediment Control Certification

Incident Command, ICS I, II, II and IV

Bridge Painting Inspections – FHA

Concrete plant inspection and certification - NYSDOT

Project Concrete – NYSDOT

Culvert Design – FHA

Earthwork Inspections – NYSDOT, Soil Mechanics Bureau

Asphalt Pavement Design and Inspections – New York Construction Materials Association

Design Considerations using FRP Composites-NYS Society of Professional Engineers

AASHTO Roadside Design – FHA

Highway Design Manual Training – NYSDOT DQAB

SPDES –Phase I and II Design Procedures- NYSDOT

Guiderail Design – NYSDOT

Pavement Design – NYSDOT

Design Procedure Manual Training – NYSDOT/FHA

Professional Societies/Affiliates

New York State Association of Transportation Engineers

Erie-Niagara Chapter of The New York State Society of Professional Engineers

American Bridge Construction and Design (ABCD) – Member

New York State County Highway Association – Member

PROFESSIONAL PROFILE

Mr. Lydle is a New York State Professional Engineer with more than years of engineering and management experience. He is a proven leader and innovator with the ability to implement and progress projects with his extensive background in planning, programming, design, construction, operations, facility management, program development, budget preparation and administration. In addition to Mr. Lydle's experience in infrastructure improvements, he is highly experienced with design, inspection, maintenance, and operation of facilities for the New York State Department of Transportation, ensuring compliance with New York State Office of General Services procedures, standards and practices and the New York State Building Codes. He is familiar with the protocols and procedures for compliance with State and Federal Regulations. Additionally, Mr. Lydle also has more than 20 years' experience providing private design and inspection services on commercial, residential and site plan designs and development for numerous projects.

EXPERIENCE

Roswell Park Cancer Institute sponsored project, Mount Aaron Community Center Improvements, Genesee Street, Buffalo NY - Construction Manager/Project Engineer – Roswell Park Cancer Institute (RPCI) in recognition of the community importance has financed and progressed community outreach building improvement projects in the City of Buffalo. LiRo Engineers, who has a long history of providing Construction/Project Management (CM) services at RPCI, is providing (CM) services on this outreach project encompassing permit acquisition, onsite inspection of all building trades, submittal reviews, and payment applications to the completion of the improvement project.

Dona Street Extension, City of Lackawanna, NY, Senior Engineer/Project Manager – The extension is an Industrial access road providing access from NY Route 5 (Hamburg Turnpike) to the former Bethlehem Steel property bordering Lake Erie. The work involved the design and construction management of this facility crossing railroad service lines and through mitigated brown field sites.

Buffalo and Fort Erie Public Bridge Authority, VACIS Relocation and Plaza Modifications, NY, Project Engineer/Manager – Provided on-site project management and inspection services for the construction of new infrastructure improvements and inspection facilities in the US Plaza of the Peace Bridge. Scope included construction of new homeland security and Customs inspection facilities and driver booths and infrastructure improvement to the plaza. New security services and equipment with fiber optics were components of this sensitive project.

City of Buffalo Department of Public Works, City-wide Island Curb Replacement, Buffalo, NY, Project Engineer - Mr. Lydle provided design and construction inspection and management for the replacement of curbs surrounding islands on various streets in the City of Buffalo. The work included Penhurst Parkway (Forest Avenue to the north end of the street), Bedford Avenue (Elmwood Avenue to Lincoln Avenue), Fordham Drive (Elmwood Avenue to Lincoln Avenue), Fordham Drive (Elmwood Avenue to Lincoln Avenue) and Viola Park Street (Pansy Place to Daisy Street). The work also included for Culver Road, Ridgewood Road, and Harding Road (all between South Park Avenue and McKinley Parkway).

City of Buffalo Department of Public Works, Sidewalk Inspection Three-Year Term Contract, NY, Project Engineer - LiRo provided engineering construction administration and inspection services for various infrastructure improvement projects for the City of Buffalo for the 2015-2017 term. The type of work included pavement construction, sidewalk construction, curb construction and other miscellaneous work within the City right-of-way. Inspectors were required to provide full-time on-site observation, reporting and coordination in accordance with DPW standards.

Michael Lydle, PE

Project Manager

Dormitory Authority State of New York, Construction Management On-Call at Niagara Falls State Park, NY, Senior Engineer - LiRo is providing construction management services for the second phase of renovation work at Prospect Point and the Lower Grove Trails within Niagara Falls State Park. The work at this popular viewing area takes place in phases to permit continual visitor access to the falls and surrounding attractions. The work includes landscape restoration, new railings, benches, light fixtures, and walkway surfaces. LiRo is also coordinating trail renovation work at the North Shore Trail and the Luna Bridge, as well as demolition and fitout work at the Kenan-Castellani Art Gallery located at De Veaux Woods. An additional 17 phases of work are anticipated throughout the Park. ($20 million)

Buffalo and Fort Erie Public Bridge Authority – United State Approach Widening Project, Phase II, NY. Resident Engineer: Interim Project Manager and Resident Engineer for the construction administration and inspection services throughout Phase II of the US Approach Widening Projects. Phase II consisted of modifications to the existing concrete vaults and abutments, completion of the piers, erection of new structural steel, placement of new concrete deck and sidewalk for the bridge widening, construction of architectural concrete retaining wall/rail, installation of new light gantries and modification to the electrical components and improvement of a new drainage system with pumps. Additional QA services provided at bearing manufacturing facility for dynamic load testing compliance verification and certification.

Erie County Department of Public Works, 2014 Bridge Painting Program, NY, Project Engineer - Responsible for management and coordination of the 2014 maintenance bridge painting program. The scope of these projects included field evaluation of the bridges; preparation of all required federal aid documentation; coordination with SHPO for historical bridges; assessment of maintenance and protection of traffic needs during painting operations; documentation for disposal of hazardous waste (lead); preparation of a Contract Award Documentation package; construction cost and quantity estimating; preparation of final specifications and cost-estimate; review of Class A Containment submittals; and construction inspection and support services.

Erie County Department of Public Works, Erie County Correctional Facility Parking Lot and Access Road Reconstruction, NY, Project Engineer - Reconstruction and improvement of the existing asphalt parking lots and access road at the Erie County Correctional Facility. The project is being executed in two phases, the first phase completed in 2015, addressed the parking lot on the north side of the site and the second phase addressing the access road on the campus once funds become available. The scope of work included complete removal of the existing asphalt to the sub-base and installation of new asphalt, new storm water drainage design, curb repairs, stripping and signage.

City of Buffalo Department of Public Works, Durham Avenue Reconstruction, Between Litchfield Avenue and Sussex Street, NY, Project Engineer - Mr. Lydle provided the detail design along with construction inspection and management services. The design was consistent with the social character of this established residential neighborhood by retaining its red brick pavement while addressing pavement deficiencies, improve safety and mobility, and replacing hazardous sidewalks sections that included ADA compliant sidewalks and ramps.

East Lovejoy Street, Clinton Street, and East Delavan Street – Streetscape Improvements from Bailey Avenue to the east City Line, Buffalo, NY: Project Engineer for the design of streetscape improvements to existing City streets. The project included streetlight replacement with LED lights, replacement of sidewalks and curb, installation of new ADA-compliant sidewalk ramps, sign replacement, milling and paving, and striping. Completed roadway lighting analysis for placement of new lights. Completed field assessment of existing conditions and provided recommendations for safety upgrades. Prepared bid documents and engineer’s estimate.

Bruce Przybyl

Firing Range Design Manager

Education

M.A., Geological Sciences, State University of New York at Buffalo, 1988

Certificate in Computer Programming, SUNY at Buffalo, 1987

B.A., Geological Sciences, State University of New York at Buffalo, 1985

Licenses/Registrations

Professional Geologist – New York

Certifications

OSHA 30 Certified

OSHA 10 Certified

PROFESSIONAL PROFILE

Mr. Przybyl is a Project Manager with extensive experience in managing complex projects including several for NYS Public and Federal agencies. For the past 5 years, he has been coordinating shooting range remediation and range design projects for LiRo. He has 30 years' varied experience in environmental remediation, engineering & construction consulting.

EXPERIENCE

Sing Sing Correctional Facility, Outdoor and Indoor Firing Ranges. Served as Project Manager for multiple phases of evaluation and planning for NYSOGS and DOCCS starting with evaluation of facility's existing outdoor range and finishing with design of a the new 8-lane Fully Enclosed Firing Range. This replacement facility can accommodate up to 100-yard AR-15 qualifying in a fully enclosed environment. The evaluation of the existing range concluded that most of the range components had reached their serviceable lifespan and LiRo recommended removal and replacement. The new indoor range, is located in a tight space, constrained by bedrock outcrops, a 70-foot high cliff, and the perimeter road and security wall. LiRo overcame the space constraints to meet all range, Preparation Suite classroom, access, utilities, and parking requirements by using the terrain as an advantage - stacking the classroom and covered parking below the Ready Area end of the range. The facility is equipped with state-of-the art ventilation and acoustical controls. Construction is underway.

Beerston Police Precinct, Firing Range Evaluation and Renovation. Walton, New York. Managed preparation of an Alternatives Analysis Report of various options to reclaim lead projectiles from this outdoor range and develop a new backstop system. Based on the analysis, prepared the design, including plans and specifications for the reclamation and new granular rubber bullet trap backstops for each of the Small Arms and .50 caliber sniper ranges to replace the existing soil berm backstops. The project was constructed in 2018.

Correctional Officer Training Academy, Range and Training Building, Coxsackie, New York. Managed the preparation of a Program Report for NYSOGS and DOCCS including code review for a new firearms training facility at the Coxsackie CF. Permitting included wetlands evaluation, archeological permitting, and SEQR Forms. Managed design and coordination of bid document preparation of an outdoor firing range and training building at the site. The facility consists of a 5,000 square foot class room training building, a shooter's shelter, a steel escalator bullet trap, a 16-station 100-yard baffled firing range with shooting positions at 100, 75-, 50-, 25-, 15-, 7- and 3-yards for AR-15 rifle, 9mm pistol, and shotgun. The range is fully baffled (no blue sky) from 25-yards to target. The facility also includes a separate CS-gas training building, and the main training building includes a heated outdoor access corridor with 6-eywash stations to support 4-season capsicum spray training. Bid documents are in preparation and construction startup is scheduled for Fall 2019.

Auburn Correctional Facility - Town of Sennett, New York. Project Manager for design of an 8-lane 100-yard Indoor Firing Range for the Auburn CF. In response to resident concerns regarding range noise impact, LiRo collected background sound data, terrain, and ground cover information and modelled sound propagation from the range using the CADNA 3-D analytical model. The range design is based on the Sing Sing CF range template fit to the terrain, view shed, and facility requirements. Construction is underway.

Amherst Police Department Indoor Shooting Range, New York. Firing Range Design Manager for a six lane 65-yard indoor tactical pistol and rifle range within the Department's multi-function law enforcement facility renovation. The project includes evaluation of space development for 270-degree live fire training. The project is in the schematic design phase.

Bruce Przybyl

Firing Range Design Manager

Firing Range Remediation, Butler Correctional Facility, Red Creek, New York. Project Manager for the remediation design of lead-contaminated soil to below NYCRR Part 375 Unrestricted Soil Clean-up Objectives. Construction was completed in the summer of 2016.

Firing Range Remediation, Mid-Orange Correctional Facility. Warwick, New York: Design Leader for the remediation of lead-contaminated soil to below NYCRR Part 375 Restricted Residential Soil Clean-up Objectives. The project was completed for the NYSOGS. The work elements included: excavation of two soil berms and remediation of the range floor, processing to remove/recycle lead shot, characterization, separation, transport of soil off-site, and disposal in accordance with applicable local, State and Federal regulations. Design included preparing a Basis of Design Report, Joint Application for Permit, Plans, and Specifications. Also managed LiRo's construction management services for the remediation completed in 2014.

Firing Range Remediation, Camp Georgetown Correctional Facility. Design Leader for the remediation of lead-contaminated soil to below NYCRR Part 375 Restricted Residential Soil Clean-up Objectives. The project was completed for the NYSOGS. The work elements included: excavation of lead contaminated soil and disposal in accordance with applicable local, State and Federal regulations. Design included preparing a Basis of Design Report, Plans, and Specifications. Also managed LiRo's construction management services for the remediation completed in 2014.

Episcopal Church Home Abatement and Demolition, Buffalo, NY: Design Manager for the abatement, demolition, and site stabilization of a former geriatric complex on behalf of the Empire State Development Corporation. Managed the development of Plan and Specifications, prepared the Environmental Assessment form, participated in public forums, and prepared community news letters. Acted as the owners engineer during abatement/demolition, completed in 2015.

Performance Based Remediation, Loring and Pease Air Force Bases: Project Manager for a 6-year performance based delivery order to perform remedial design & construction, operations, maintenance, and monitoring (OM&M), and site source closure at two closed BRAC bases. At Pease AFB in New Hampshire, the team executed design-build remediation projects including:

- A large-scale air sparging system (Plume 13/14). The site is situated in the main taxiway; a temporary taxiway was constructed to enable uninterrupted airfield operations during the construction period. Agreement on a final design required extensive coordination with the Pease Development Authority (the airport managing authority) and the New Hampshire Department of Environmental Services (NHDES). Installing the system involved demolishing airport pavement, installing the system below grade (which included nearly 2 linear miles of piping), and replacing the pavement per FAA guidelines.

- A dig and haul (Plume 41). The team determined that site source closure could not be achieved during the contract period by continuing operation of the exiting remedial system. Therefore, excavation and disposal of the soil source was executed and site closure approved. To execute the removal, the team first demolished the taxiway pavement, excavated the contaminated soil, collected confirmation analytical samples, disposed the excavated soil as non-hazardous waste, and restored the taxiway with 14-inches of flexible pavement per FAA guidelines.

At Loring AFB in Maine, design-build remediation projects included:

- At the Fuel Tank Farm, the team demolished abandoned infrastructure, excavated 29,000 CY of fuel-contaminated soil, remediated it to Maine Department of Environmental Protection (MEDEP) satisfaction at landfarm cells constructed on the inactive PLS runway, and replaced the remediated soil into the excavation. The schedule was compressed because of the presence of a State threatened avian species nesting adjacent to the runway.

- At NDA 2, the team successfully remediated 30,000 CY of contaminated soil in 2008. To avoid a construction season shortened by the presence of a State threatened avian species nesting adjacent to the existing landfarm cells, the team designed and constructed a new landfarm able to contain runoff from a 100-year flood.

Rochester Gas and Electric (RG&E) Beebee and Russell Power Station Demolition and Remediation Design, Rochester NY: Senior Scientist supporting design of demolition and remediation of the retired Beebee Station electric generating station located on the bank of the Genesee River and the retired Russell Station coal-fired electric generating station located at the Southern shore of Lake Ontario. In addition, conducted an adaptive reuse study of the Beebee Station. The design also included protection of existing switch yard and transmission infrastructure, utility terminations, tunnel closures, access road repairs, and site restoration.

Madawaska Dam Treatment Plant Upgrade, Caribou, ME: Project Manager for upgrade of the treatment facility supplying potable water to the Loring Commerce Park at the former Loring Air Force Base. The facility once supplied 2,000,000 gallons per day (gpd) of potable water to this former Strategic Air Command base prior to base closure. The facility currently fills the water needs of the commerce park, drawing about 150,000 gpd with a potential need of over 500,000 gpd as the park fills with commercial tenants. However, the water system was sporadically out of compliance with recent EPA rules for chlorination byproducts. The project team studied, designed, and executed upgrades to this surface water treatment facility, successfully bringing the potable water system into regulatory compliance for chlorination byproducts. The project team prepared design drawings and specs for the improvements, solicited contractors to execute the various upgrade elements, and supervised construction.

Multi-Media Sampling - Niagara Falls Air Reserve Station, NY: Program Manager for a performance based indefinite delivery/indefinite quantity (IDIQ) task order contract for monitoring, sampling, analysis, and reporting for various environmental media across this active base. Managed 65 task orders for monitoring of oil-water separators, monitoring of sanitary sewer flow and discharge quality, inspection of storm water flow, asbestos inspection, and sampling of buildings slated for renovation, asbestos abatement monitoring, and monitoring various waste streams on base to characterize the waste streams for disposal in accordance with State and Federal regulations.

Former EMCA Site, Mamaroneck, New York: Project Manager for actions at this former Dow Chemical Company site to address Freon-113-contaminated groundwater. The selected cleanup technology, injection of a mixture of vegetable oil and sodium lactate, was first implemented as a successful pilot study. This study was followed-up using the same technology in an Interim Remedial Action (IRA) and three supplemental injections. The latest event included bioaugmentation using dehalococcoides and dehalobacter. The injections resulted in a two to three order reduction in Freon 113 concentration and the sequential production and reduction of its degradation products. Documents prepared included an Engineering Evaluation/Cost Analysis (EE/CA), Pilot Study Work Plan and Report, HASP, IRA Work Plan, Site Management Plan (SMP), and Environmental Easement. Management of the site included sub-slab and indoor air sampling to evaluate potential vapor intrusion into the on-site building, which serves as an active cable company service center.

Richard Blanch, PE

Senior Mechanical Engineer / Acoustical Engineering

Education
B.S., Mechanical Engineering, The State University of New York at Buffalo

Licenses/Registrations
Professional Engineer, New York

Certifications
OSHA 10 Certified

PROFESSIONAL PROFILE
Mr. Blanch has over 30 years of extensive project management experience for the design and construction of private, commercial, healthcare, municipal, and government agency projects. Experience includes the design of HVAC systems, plumbing, ventilation systems, medical gas systems, noise impact and acoustical remediation, water and wastewater plant design, and process cooling systems. Mr. Blanch has managed design and construction teams, including construction phase coordination between contractors and understands the importance of detailed project scoping for both the design team and the client.

EXPERIENCE
New York State Office of General Services, Firing Range Design & Rehabilitation New York Statewide, NY – Under this task order contract, thus far 3 projects have been awarded totaling $1,000,000 for design of one outdoor and two indoor correctional facility shooting ranges. The design of each range required meeting ballistic containment and noise reduction objectives. *Mr. Blanch is the MEP Project Manager for the OGS Firing Range Term Contracts. Rick coordinates all aspects of integrated mechanical and electrical systems with the project stakeholders, to ensure the project objectives are met, and with consideration for range operations, ease of maintenance, and energy. Rick also led acoustical modelling for all 3 projects to date using the CADNA 3-D model, which assured the potentially affected communities that the sound of gunfire would not cause adverse impact.*

Correctional Officer Training Academy Firing Range and Training Building - Coxsackie, NY. LiRo completed design of an outdoor firing range and training building for the Albany Correctional Officer Training Academy at the Coxsackie Correctional facility in July 2017. The facility consists of a 5,000 square foot class room training building, a shooter's shelter, a steel escalator bullet trap, a 16-station 100-yard baffled firing range with shooting positions at 100-, 75-, 50-, 25-, 15-, 9- and 3-yards for AR-15 rifle, 9mm pistol, and shotgun. The range is fully baffled (no blue sky) from 25-yards to target. The facility also includes a separate CS-gas training building, and the main training building includes a heated outdoor access corridor with 6-eywash stations to support 4-season capsicum spray training. Bid documents are in preparation and construction is scheduled for 2020.

Sing Sing Correctional Facility Firing Range - Ossining, NY. LiRo designed the new Sing Sing CF 8-lane Fully Enclosed Firing Range. This facility can accommodate up to 100-yard AR-15 qualifying in a fully enclosed environment. The site is located in a tight space, constrained by bedrock outcrops, a 70-foot high cliff, and the perimeter road and security wall. LiRo overcame the space constraints to meet all range, Preparation Suite classroom, access, utilities, and parking requirements by using the terrain as an advantage - stacking the classroom and covered parking below the Ready Area end of the range. The facility is equipped with state-of-the art ventilation and acoustical controls. Electrical and natural gas connections are designed separate from the main correctional facility utilities with local backup to enable uninterrupted service. The facility is equipped with an interlocking range control system and data is provided linking the security and mechanical systems to the main correctional facility information system. Construction is underway.

Auburn Correctional Facility - Town of Sennett, NY. LiRo has completed the design of an 8-lane 100-yard Indoor Firing Range for the Auburn CF. In response to resident concerns regarding range noise impact, LiRo collected background sound data, terrain, and ground cover information and modelled sound propagation from the range using the CADNA 3-D analytical model. The range design is based on the Sing Sing CF range template fit to the terrain, view shed, and facility requirements. Construction started in Spring 2019.

Richard Blanch, PE

Senior Mechanical Engineer

Federal Aviation Administration, Noise Abatement Projects at Various Airports, Mechanical Engineer - Mr. Blanch was responsible for field data collection, acoustical analysis, and technical writing. He was also part of the interface team which met with the residential and commercial clients within the impacted noise contours. Noise impact studies and noise mitigation designs were generated for runway expansions at the Syracuse, Islip, Pittsburg and Cleveland airports were studied. The acoustical treatments included mechanical system modifications for commercial and residential structures. The noise studies were repeated after abatement and runway projects were completed to verify remediation and acceptable acoustical results.

National Fuel Gas, Noise Impact Study for Compressor Stations, Western New York State, Mechanical Engineer - Free field noise levels were measured around the compressor stations and impact levels for the compressor upgrades were calculated. Attenuation alternatives where noise levels were projected to exceed local ordinances were addressed in a final report. Mr. Blanch was responsible for field data collection, as well as performing the ambient noise impact calculations and compiling into the final report.

Orleans County, Landfill Noise Impact Study, Albion, NY, Mechanical Engineer - A noise impact study was conducted for traffic and heavy equipment related to proposed incremental noise for the landfill construction site. Field data collection, as well as all calculations and report writing for the noise study, were included. Mr. Blanch was responsible for the calculations, field data collection, as well as the technical report writing and presentation.

NYS OGS - Troop C Headquarters Renovations S5178, Unadilla, NY - The design comprised new mechanical and electrical systems associated with the Forensic Investigation Unit (FIU) at a 9,000 sf State Police Headquarters building in Unadilla, NY. The design included mechanical systems design of an indoor firing range. The medium pressure duct systems provided laminar flow and filtration to minimize risk of toxic lead exposure. Mr. Blanch was the Sr. Mechanical Engineer responsible for quality review and assurance.

US Army - Fort Drum Jet Refueling Facility, Watertown, NY - Mr. Blanch was responsible for the design of the ventilation, heating, and environmental control systems for the jet storage and refueling hangars.

NFTA - BNIA Security Screening Area Modifications, Cheektowaga, NY - Mr. Blanch was the Project Manager responsible for the overall project coordination of MEP design and construction administration and inspection services. The project involved mechanical and electrical infrastructure modifications for the TSA analogic passenger scanners and the adjacent concessions and retail space reprogramming. The design required capacity and efficiency studies to effectively condition and power the security and concessions equipment.

NFTA - BNIA TPAO HVAC Rehabilitation, Cheektowaga, NY - Mr. Blanch was the Project Manager responsible for overall coordination and project management of the design team. The design modifications included upgrade to the Building Management control systems to current technology.

Westchester County Department of Public Works, New Infirmary Building for Westchester Correction Department, NY, MEP Engineer - LiRo is providing architectural and engineering design services for a new infirmary building at the Valhalla Campus at Grasslands. The infirmary facility will consist of two buildings: the infirmary treatment facility and the housing unit, totaling approximately 13,500 sf and will include medical examination and treatment areas. The new unit will also include several isolation suites, a day room for infirmed inmates and operation areas for support staff. A new suicide watch suite building, which is approximately 3,000 sf, will be situated adjacent to the infirmary and will be served by the staff support spaces within the infirmary. ($16 million)

Richard Blanch, PE

Senior Mechanical Engineer

NYS OGS - New York Education Department- Data Center Renovations, Albany, NY - The project included critical environment HVAC and fire protection systems for a new raised floor computer center for the New York State Education Department. The HVAC systems consists of multiple chilled water computer grade A/C units that provide approximately 100 tons of normal and emergency cooling for the facility. The fire protection system is a total flooding FM 2000 system that protects the above and below raised floor space. Mr. Blanch was the Mechanical Engineer and Project Manager responsible for mechanical systems design and project coordination with the prime architect and owner for this project.

NYSOGS - Taconic Correctional Facility- Clinic Expansion, Bedford Hills, NY - This project involved surgical renovation and a 3000sf addition to the Medical Services Building on the Taconic Correctional Campus. The existing building was a wood framed single story pre-manufactured building, which posed challenges for the new addition with respect to mechanical, electrical, plumbing, and fire protection infrastructure. Mr. Blanch was the Project Manager responsible for design team coordination and client interface.

NYSOGS -Woodbourne Correctional, Building 4A, Woodbourne, NY - Design challenges included a construction phasing plan to abate the environmental hazards within the building, and strategically cost effective mechanical and electrical modifications and enhancements. Mr. Blanch was the Project Manager and Mechanical Engineer responsible for the program reports and MEP design.

NFTA - Buffalo Niagara International Airport- New ARFF Facility 31/32BR 1406, Cheektowaga, NY - A new 27,000sf air rescue and fire-fighting facility at the Buffalo Niagara International Airport was designed for the Niagara Frontier Transportation Authority. The new ARFF will house critical IT infrastructure for operation of the NFTA Area Wide network and the NFTA phone system head end. The mechanical system design includes a variable refrigerant flow system, which has the ability to provide simultaneous heating and cooling (prevalent during swing season months), where the transfer of heat from one zone to another requires very little energy. The construction cost for this project was $9.4 million. Mr. Blanch was the Project Manager responsible for coordination of the mechanical, communications, and fire protection designs.

NFTA - Authority Wide- Perimeter Security & Access Control, 12AW0934, Buffalo, NY - The Niagara Frontier Transportation Authority (NFTA) contracted with U&S Services Inc. for a Design-Build procurement to provide design and construction of perimeter security at several garages, maintenance yards, offices, and the Buffalo Niagara International Airport (BNIA). New high definition and thermal imaging cameras provide upgraded video surveillance capabilities with remote command and control at both the BNIA and the Operation Control Center in downtown Buffalo. The perimeter security measures include new fencing, operable gates, fiber optic network, and access control systems. Mr. Blanch was the Project Manager responsible for design team coordination and client interface.

Buffalo Bills, Administration Offices, HVAC Systems, Orchard Park, NY, Mechanical Engineer - Mr. Blanch was responsible for equipment design and selections for the mechanical systems. A water source heat pump system, therapy quench tank cooling chiller system, and building ventilation air handling system comprised the primary aspects of the HVAC systems for the new administration offices for the Buffalo Bills. The mechanical room housed a R134a screw chiller and high-efficiency boilers.

Stephen Frank, PG

Senior Geologist

Education

B.S., Geology, Cleveland State University, 1986

Post Baccalaureate Studies, Geology, Engineering Geology, and GIS, SUNY at Buffalo, 1989

Licenses/Registrations

Professional Geologist, New York

Certifications

OSHA 40 Hazwoper Certified

OSHA 30 Certified

Underground Storage Tanks

PROFESSIONAL PROFILE

Mr. Frank is a Senior Geologist with more than 29 years' experience in site evaluations, aquifer testing, groundwater modeling, project coordination, environmental sampling, soil management, supervision of field activities, and report development at various hazardous waste sites, industrial sites, and commercial sites. He has extensive experience in working with the New York State Department of Environmental Conservation (NYSDEC) under various brownfield and consent order programs and working on New York City (NYC) re-development projects under the City Environmental Quality Review (CEQR) program. He also has specialized experience coordinating field investigation, reporting, and design tasks for multiple-site work order based contracts.

EXPERIENCE

Erie Canal Harbor Development Corporation, Buffalo Canal Side Development, NY - LiRo is providing construction management services for infrastructure expansion, canal extensions, and pedestrian bridges, which will form the foundation for a planned mixed-use redevelopment of this historic waterfront site. The Canal Side project is a redevelopment program that is important to restoring the City of Buffalo's waterfront vitality. The project area covers approximately 20 acres of idle waterfront land. The master plan proposes nearly 725,000 sf of mixed-use space for entertainment, hotel, office, retail, residential, restaurant, and other uses. ($315 million)

1318 Niagara Street Site, Environmental Restoration Program, Buffalo, NY: Project Coordinator responsible for Interim Remedial Measure (IRM) and environmental subsurface investigations at a contaminated former industrial site in the City of Buffalo, NY. Responsible for development of IRM and subsurface investigation work plans, coordination of IRM subcontracting for the removal of petroleum and PCB-contaminated soil, removal of drummed hazardous waste, cleaning and removal of former underground storage tanks (USTs), performing subsurface sampling, and interpretation of subsurface conditions and environmental sampling results. Mr. Frank will also lead the Project remedial design team effort, develop a remedial Alternatives Analysis Report, and assist the New York State Department of Environmental Conservation (NYSDEC) in the preparation of a Proposed Remedial Action Plan (RAP) for the site.

Dormitory Authority State of New York, Construction Management On-Call at Niagara Falls State Park, NY - LiRo is providing construction management services for the second phase of renovation work at Prospect Point and the Lower Grove Trails within Niagara Falls State Park. The work at this popular viewing area takes place in phases to permit continual visitor access to the falls and surrounding attractions. The work includes landscape restoration, new railings, benches, light fixtures, and walkway surfaces. LiRo is also coordinating trail renovation work at the North Shore Trail and the Luna Bridge, as well as demolition and fitout work at the Kenan-Castellani Art Gallery located at De Veaux Woods. An additional 17 phases of work are anticipated throughout the Park. ($20 million)

Buffalo Urban Development Corporation – Northland Corridor Redevelopment, Buffalo, NY: Senior Environmental Scientist for the Northland Corridor project, which comprises 12 properties and 50 acres of a former industrial district within the East Delavan-Grider Brownfield Opportunity Area in Buffalo, NY. LiRo was contracted by the Buffalo Urban Development Corporation to formulate a redevelopment plan for acquiring multiple properties that will eventually lead to returning the Northland Corridor to productive use, attract new businesses, assist with revitalizing the surrounding neighborhood, and provide employment opportunities for nearby residents by creating a new manufacturing hub on Buffalo's East Side. The project includes real estate/ market analysis, SEQR documentation, historical preservation, planning/ transportation/ infrastructure analysis, remediation cost estimating and community participation to culminate in a detailed Redevelopment Plan.

Stephen Frank, PG

Senior Geologist

Erie Canal Harbor Development Corporation – Buffalo Outer Harbor Parcel OH Site Improvements: Project Principal for the redevelopment design of two parcels on Fuhrmann Boulevard located on the Outer Harbor in Buffalo, New York. The project consists of preparing approximately 15 acres for future development along with providing amenities for public access. The work included shoreline enhancements, public walkways and bike paths, water taxi docking facilities, landscape features, parking, site lighting and the rehabilitation of an existing metal building for use as a comfort station and storage building. Landscape features included rain gardens and retention ponds for runoff. Previous site usage and contamination required placement of an earthen cap over the entire site and removal of an underground petroleum storage tank and the associated piping.

Erie Canal Harbor Development Corporation (ECHDC) Buffalo Memorial Auditorium (AUD) and Donovan Office Building (DOB) Construction Management Services, Buffalo, NY: Regulatory Specialist for construction management services of environmental projects for the AUD and DOB. LiRo has was retained by the ECHDC to perform construction management services to coordinate and oversee asbestos abatement and environmental remediation work at the AUD and DOB in downtown Buffalo, NY. The AUD is a 200,000+ square foot (S.F.) arena that was constructed in 1939. The DOB is a 160,000 S.F. eight-story office building. The AUD and DOB properties were the proposed site for a mixed-use retail complex including structured public parking facilities. The objective for this project was to perform all asbestos and hazardous materials abatement to allow for building demolition under a subsequent contract. The performance of the asbestos abatement and hazardous materials removal work under the proposed project schedule of 18 months was a critical element of the overall project schedule for demolition and site redevelopment.

Spaulding Fibre Plant Environmental Restoration Program, Tonawanda, NY: Project Coordinator responsible for environmental subsurface investigations at a 47-acre former industrial facility. Responsible for development of subsurface investigation work plans, coordination of subcontractor activities, performing subsurface sampling, and interpretation of subsurface conditions and environmental sampling results. Mr. Frank also lead the Project remedial design team effort, developed remedial plans and specifications, provided bid support services, reviewed Contractor submittals, and is coordinating the site inspection for a $3 million site remediation/foundation demolition program.

RG&E Beebee and Russell Power Station Demolition and Remediation Design, Rochester NY: Environmental Project Manager for the demolition and remediation of the retired Beebee Station electric generating station, located on the bank of the Genesee River, and the retired Russell Station coal-fired electric generating station, located at the Southern shore of Lake Ontario. In addition, LiRo conducted an adaptive reuse study of the Beebee Station. The design also included protection of existing switch yard and transmission infrastructure, utility terminations, tunnel closures, access road repairs and site restoration.

Christopher Abramo, PE

HVAC Engineer

Education

B.S., Mechanical Engineering, Rochester Institute of Technology

Licenses/Registrations

Professional Engineer, New York

Professional Engineer, Nebraska

Professional Engineer, Virginia

Professional Engineer, Georgia

LEED AP BD+C

PROFESSIONAL PROFILE

Mr. Abramo has over 13 years of experience in project management and mechanical engineering design. His experiences include heating, ventilating, and air conditioning systems for new and existing buildings. Assignments have included the design and construction of private, commercial, healthcare, municipal, transportation, and government agency projects. Mr. Abramo has experience in project management, having led design and construction teams, including managing project schedules, budgets, and coordination with contractors.

EXPERIENCE

DASNY At-Risk Buffalo State College Tower 3 Renovations - LiRo is providing, on an at-risk basis, professional construction management services during the design and construction of Tower 3, a 10-story dormitory with 212 beds on the campus of Buffalo State College. The scope of this project will be to renovate all 10 floors and the basement, with LEED Silver Certification.

Amherst Police Department Indoor Shooting Range, New York. Project Manager and lead HVAC Engineer for a six lane 65-yard indoor tactical pistol and rifle range within the Department's multi-function law enforcement facility renovation. The project is in the schematic design phase.

New York State Office of General Services, Firing Range Design & Rehabilitation New York Statewide, NY – Chris served as HVAC and controls designer on the for two indoor correctional facility shooting ranges.

Sing Sing Correctional Facility Firing Range - Ossining, NY. LiRo completed design of the new Sing Sing CF 8-lane Fully Enclosed Firing Range. This facility can accommodate up to 100-yard AR-15 qualifying in a fully enclosed environment. The site is located in a tight space, constrained by bedrock outcrops, a 70-foot high cliff, and the perimeter road and security wall. LiRo overcame the space constraints to meet all range, Preparation Suite classroom, access, utilities, and parking requirements by using the terrain as an advantage - stacking the classroom and covered parking below the Ready Area end of the range. The facility is equipped with state-of-the art ventilation and acoustical controls. Electrical and natural gas connections are designed separate from the main correctional facility utilities with local backup to enable uninterrupted service. The facility is equipped with an interlocking range control system and data is provided linking the security and mechanical systems to the main correctional facility information system. Construction is underway and substantially complete.

Auburn Correctional Facility - Town of Sennett, NY. LiRo has completed design of an 8-lane 100-yard Indoor Firing Range for the Auburn CF. In response to resident concerns regarding range noise impact, LiRo collected background sound data, terrain, and ground cover information and modelled sound propagation from the range using the CADNA 3-D analytical model. The range design is based on the Sing Sing CF range template fit to the terrain, view shed, and facility requirements. Construction started in Spring 2019.

New York Power Authority, Fresh Pond Bus Depot HVAC Upgrade, NY, Mechanical Engineer - LiRo is currently providing NYPA energy engineering consulting and construction management services in the South-East New York state territory in connection with energy efficiency improvement projects affecting building structures and systems. HVAC Equipment Replacement at Fresh Pond Bus Depot includes replacing all rooftop HVAC equipment with new high efficient units. The 213,000 sf facility includes both packaged DX rooftop units and split system CRAC units serving critical electrical and communications rooms. LiRo's scope included payback analysis, cost estimating, and HVAC design of replacement systems.

Connecticut Department of Transportation, Waterbury Maintenance Facility, Waterbury, CT, Lead Mechanical Engineer - Design of the maintenance related equipment, storage equipment, and specialty fluid distribution design. The design is for a new 275,000 sf facility to house 50 full size buses and 50 paratransit vehicles, including parking, service, administration, and vehicle storage. Fluid distribution included the design of a gasoline dispensing and canopy system and a bulk distribution of diesel fueling within the building. Additional deign items included new bus lifts, vehicles wash system, paint spray booth, parts storage, maintenance service pit, and tire storage. (2012- 2017)

Erie County Water District, Warehouse Study, Lead Mechanical Engineer - Design of the HVAC and storage equipment layout to optimize the storage, safety, and functionality of the warehouse space. Mr Abramo was responsible for the layout and selection of storage equipment within the warehouse and helping with the space circulation within the warehouse. Additionally, the study addressed HVAC upgrades to provide adequate heating and ventilation.

Niagara Frontier Transportation Authority (NFTA), Consolidation Study, Buffalo, NY - Performed a state of good repair and operational assessment for eight of their facilities, three of which are bus maintenance garages. (2015-2016)

Greater Lynchburg Transit Company (GLTC), Kemper St. Transfer Facility, Lynchburg, VA, Lead Mechanical Engineer - Design of a 52,000 sf maintenance, service, and administration building. Design responsibilities included heating, ventilating, and cooling, fuel and fluid distribution, and maintenance equipment and storage equipment layout and selection. The building HVAC system included two large gas fired heat recovery units that provided the required ventilation to that garage, two air handing units to heat and cool the administration building, and a central hot water distribution system. Additionally, the facility was designed to accommodate service of CNG vehicles. A gas detection system controlled emergency exhaust fans to accommodate service of these vehicles. The fueling system design included one 15,000 gallon diesel fuel tank, one 10,000 gallon gasoline tank, an exterior gasoline fuel lane and canopy, and an indoor diesel fuel lane with a tramway fueling system. Fluid distribution included overhead fluid reels, multiple indoor fluid storage tanks, fuel and fluid management system, waste oil and waste coolant pump out, etc.

Alfred University, Inamori Museum of Ceramics, Alfred, NY, Mechanical Engineer - Responsible for HVAC and plumbing design of new ceramics museum within existing building. Open ceiling architectural design required precise system layout coordination to conceal all ductwork and piping within the museum space.

State University Construction Fund, SUNY College of Ceramics, Binns Merrill Hall Heat Recovery Phase II, Alfred, NY - Design and construction services to renovate the equipment and area associated with the Glass Blowing program. The project included architectural, structural, electrical, and mechanical engineering to support the upgrades to the existing glass blowing furnaces. The existing exhaust hood was demolished and a new hood provided to effectively capture excessive heat from the furnaces and provide energy savings through a closed-loop heat recovery system. Industrial ventilation codes and standards were used to effectively size the associated equipment, capture hoods, and ventilation system. An opinion of probable cost was provided at each design submission to verify that the project scope remained within the allocated budget during the design process. One design challenge was to relocate the ventilation system from within the space to the roof and provide connecting ductwork from the first floor hood to the roof mounted fans. Based on the opinion of probably cost it was determined that a cost saving measure was required to accommodate other project costs. A less costly alternative was developed for the duct routing and modify the bid documents to include a bid alternate. Full construction services, including running project meetings and coordination with SUCF construction coordinator, the college, and the contractor.

Buffalo State, Rockwell Hall HVAC, Lead Mechanical Engineer - Replacement of multiple air handling units providing heating, cooling, and Ventilation to the main concert hall, music rooms, choral rooms, and atrium. Two units served the existing concert hall, one indoor and one outdoor unit. Both units had separate condensing units. The project scope included removing the refrigerant based system and extending chilled water from the new central plant. The air handling units were replace with two roof top units. There was limited room on the roof for these units, structural capacity of the roof was a major consideration, and sound transmission was critical. One of the units was located over a sound recording studio and both served a concert hall where air noise would be very disruptive. Fan speed, ductwork size, insulation requirements, acoustical turning vanes, air velocities, insulated curbs, isolated curbs, and fan vibration isolation were factors considered in the final design. Additional design features included, extending the central chilled water and hot water systems to the new air handling units, demand control ventilation, upgrade of all controls, economizer cooling, and variable speed pumping. The two air handling units serving the atrium and music rooms were replaced in kind and cooling added to the atrium. Space constraints was a limiting factor and required very detailed field work and coordination.

New York State Parks, Visitor Center, HVAC Upgrade, Lead Mechanical Engineer - Design and construction for the complete rehab of the HVAC system associated with the historic Niagara Falls Visitor center. A schematic design report and estimate were presented to the client for providing the client with a budget conscious solution. One of the major design considerations was to keep the visitor center open to the public throughout construction. Temporary heating, cooling, power, occupant safety, etc were major considerations. Working closely with the client and contractors was key to providing a successful project. (2014-2016)

Veterans Affairs Western New York Health Care System, X-Ray Suite Renovation, Buffalo, NY, Mechanical Designer - Mr. Abramo was the lead mechanical engineer for the installation of a new x-ray machine. Design included extending ductwork from an existing air handling unit and modifying the existing central plant chilled water and hot water piping to accommodate the space renovations. A dedicated supplemental cooling system was installed within the X-ray room to accommodate tight temperature and humidity requirements. The Ventran Affairs Healthcare design guide was referenced as the primary design guide.

Yahoo! Data Center, Lockport, NY- Site plan approvals, permitting, field topographic and ALTA survey concept through final site, civil, mechanical and landscape design services, SWPPP, construction support and electrical, mechanical commissioning services for $150M, 115,000 sf Data Center on 30 acre parcel in the Town of Lockport IDA Park.

Frank Ulisse, PE, PMP

QA/QC

Education

B.S., Electrical Engineering, Rutgers University

Licenses/Registrations

Professional Engineer, Florida

Professional Engineer, Maryland

Professional Engineer, Pennsylvania

Professional Electrical Engineer, Massachusetts

Professional Engineer, Delaware

Professional Engineer, New York

Project Management Professional

Professional Electrical Engineer, Rhode Island

Professional Engineer, New Hampshire

Professional Engineer, Virginia

Professional Engineer, New Jersey

PROFESSIONAL PROFILE

Mr. Ulisse is experienced in senior management of multi-disciplined engineering projects. With more 30 years of experience, he has served as an executive for multidiscipline engineering projects involving the design of electrical, HVAC, plumbing, fire protection systems, civil, structural, geotechnical, and environmental disciplines. His areas of technical expertise consist of medium and low voltage power distribution, lighting, grounding, power quality, and life safety. His experience includes commercial, industrial, utility, and civil infrastructure projects, including educational facilities, healthcare facilities, office buildings, parking garages, retail spaces, hotels, casinos, municipal buildings, operable bridges, and military installations. He has extensive experience with renewable energy systems, including solar photovoltaic electrical systems. He has managed many LEED projects that have included energy efficient systems, such as ground source heat pumps (geothermal). His experience also includes managing general engineering and on-call contracts for several federal, state, and local government agencies. He also has managed large, multidiscipline, design-build projects and international projects.

EXPERIENCE

New Jersey Department of Treasury, Division of Property Management and Construction (NJDPMC), Valentine Hall Chiller Replacement in Bordentown, Burlington County, NJ, Principal-In-Charge - The Johnstone Training School, operated by the Juvenile Justice Commission, rests on a 97-acre parcel of land that includes a four-building campus listed on the National and State Historic Registers. This project involves a chiller replacement for Valentine Hall, a 27,540-square-foot, brick-façade structure built in 1929 and partially rebuilt in 1986. Scope includes providing design and specifications for a new energy-efficient electric chiller system, taking into consideration restrictions applying to security, site parameters, facility personnel, staff, and residents as the building will be occupied during the construction phase. ($195,000)

Philadelphia International Airport (PHL), American Airlines (AA) Design and Engineering Services, Terminal Modernization Program Phase 1 Extended Design, Philadelphia, PA, MEP Principal - AA and the PHL are in the process of reconfiguring and streamlining its terminal and associated support facilities. HAKS, as a team member, has been performing MEP and fire protection engineering services for the enabling projects in the terminal. These projects are part of a phased approach to constructing a new headhouse at Terminal B/C. HAKS is also providing mechanical, plumbing, and fire protection engineering for a storage warehouse adjacent to the existing American Airlines repair hangar in Cargo City. ($2 billion)

New York City Police Department, Coney Island Firing Range Reconstruction, Brooklyn, NY, Project Manager - The range was originally built in 1996 for the New York City Transit Police and has been taken over by the NYPD. It consists of four shooting bays of firing lanes on each side of a common control room. HAKS is providing engineering and design services for reconstruction of the NYPD's Coney Island firing range. Scope of work includes preparing and furnishing required contract specifications and drawings, including investigations, inspections, documentation and certification, filing and permitting needed to prepare complete and accurate bid documents for the installation of new firing range systems. ($30 million)

Guilford Truck Wash Facility for Town of Guilford, CT, MEP Principal - The project entails providing architectural/engineering services for the Town of Guilford's new truck wash facility to be located at 47 Driveway. The facility will be connected to the garage building to maximize utilization of existing utilities. Since there are no sanitary sewers in Guilford, a wastewater recycling system must be incorporated into the design; an efficient fresh water truck rinsing system is critical to curb ongoing operational costs. The design will include a fully automated washing system. ($500,000)

Frank Ulisse, PE, PMP

QA/QC

Amtrak Northeast Region A/E Design and Engineering Services, Project Director - HAKS has been contracted to perform on-call work for stations and facilities, electric traction, communication and signal and track to provide safe, efficient, and economical conditions for passengers and employees. Disciplines include architectural, mechanical and structural design; HVAC; electrical engineering; civil engineering/site surveying; construction cost estimating; rail signal design; electric traction; track/civil; and historic preservation.

Liberty Property Trust, Philadelphia Navy Yard LEED Fundamental and Enhanced Commissioning and Energy Modeling, Philadelphia, PA, MEP Principal - The Philadelphia Navy Yard is a 1,200-acre dynamic and urban development, offering the region a unique and centrally located waterfront business campus committed to smart energy innovation and sustainability. Scope of work includes LEED fundamental and enhanced commissioning and energy modeling services for a 175,000-square-foot, two-story core and shell building in the Philadelphia Navy Yard. Commissioning involves site lighting, building exterior-mounted lighting, ventilation of two stair towers, and interior code required egress lighting.

Con Edison, Security Upgrades at the Learning Center, College Point and Cleveland Street, Queens and Brooklyn, NY, MEP Principal – HAKS is preparing signed and sealed construction drawings for filing, and bidding and construction documents for all required items. The drawings include one-line diagrams, wiring diagrams, plans, elevations, sections, details, notes, and specifications. Engineering services include all disciplines required for the project. HAKS is also providing administrative support during the project's bidding and construction phases.

Naval Air Engineering Station, Lakehurst Naval Base, IDIQ On-Call Contract, Principal-In-Charge – Multidiscipline indefinite delivery indefinite quantity (IDIQ) engineering contract with the Lakehurst Naval. Completed many projects between 1994 to 2008 including major HVAC upgrade projects to facilities, replacement of the base's main primary distribution substation, design of a new warehouse, connector between Hangers 2 and 3, classroom renovations, and many other projects.

Controlled Environment in Lab Building 355 – Lakehurst Naval Base, Lakehurst, NJ, Principal-In-Charge – Project that required the development of plans and specifications for a new dedicated HVAC system to provide a controlled environment in the laboratory on a year round basis, design of two new exhaust systems for two existing hoods located in the laboratory, and design modifications to two existing interconnected exhaust hood systems in the laboratory.

Yokota Air Base, Japan, Upgrade Security Gates - Provided electrical design to support the implementation of gate security measures to upgrade force protection at the six entry points to the Yokota Air Base. Design included providing power distribution and control wiring for various equipment including hydraulically operated rising road barricades, motor operated tire treadles, under vehicle inspection cameras, pan-tilt-zoom security cameras, upgraded lighting, and new guard house facilities. Work included design of new medium voltage unit substations to create dedicated secondary distribution for the new equipment. Underground conduit and manhole systems were designed for interconnection of power and control circuits to new equipment.

Burlington Federal Building, Burlington, VT, Principal-in-Charge – The energy auditing and retro-commissioning of a 148,000 square foot federal building. Reviewed record drawings and building automation system (BAS) control sequences to develop a functional performance testing plan for the retro-commissioning of HVAC and BAS systems. Conducted retro-commissioning of equipment and systems located at the facility to confirm operation conforms to the original design intent. Prepared report detailing observations and identifying deficiencies. Identified numerous potential energy conservation measures (ECM) to be considered. Prepared construction documents to implement select ECM's.

Frank Ulisse, PE, PMP

QA/QC

Warren B. Rudman US Courthouse, Concord, NH, Principal-in-Charge – The energy auditing and retro-commissioning of a 200,000 square foot federal courthouse. Reviewed record drawings and building automation system (BAS) control sequences to develop a functional performance testing plan for the retro-commissioning of HVAC and BAS systems. Conducted retro-commissioning of equipment and systems located at the facility to confirm that the operation conforms to the original design intent. Prepared report detailing observations and identifying deficiencies. Identified numerous potential energy conservation measures to be considered for implementation.

Cleveland Federal Building, Concord, NH. Principal-in-Charge – The energy auditing and retro-commissioning of a 123,000 square foot federal building. Reviewed record drawings and building automation system (BAS) control sequences to develop a functional performance testing plan for the retro-commissioning of HVAC and BAS systems. Conducted retro-commissioning of equipment and systems located at the facility to confirm operation conforms to the original design intent. Prepared report detailing observations and identifying deficiencies. Identified numerous potential energy conservation measures to be considered for implementation.

Edward T. Gignoux Courthouse, Portland, ME. Principal-in-Charge – The energy auditing and retro-commissioning of a 86,000 square foot federal courthouse. Reviewed record drawings and building automation system (BAS) control sequences to develop a functional performance testing plan for the retro-commissioning of HVAC and BAS systems. Conducted retro-commissioning of equipment and systems located at the facility to confirm operation conforms to the original design intent. Prepared report detailing observations and identifying deficiencies. Identified numerous potential energy conservation measures to be considered for implementation.

Bensalem Township Municipal Building Additions and Renovations, Bensalem, PA, Principal-in-Charge – Mechanical, electrical, plumbing, fire protection, and sustainable design engineering for the additions and renovations to the Bensalem Township Municipal Building. The new addition, 13,500 SF, and building renovations, approximately 9,000 SF were for the Police Facilities wing and included minor renovations to the Fire Marshall's offices in the Administrative wing. An addition (1,500 SF) to the detached Auxiliary Garage was also included. The base HVAC system design included the upgrade and modification to the existing water source heat pump (WSHP) system. The major components of the design were the provision of a new, closed loop, cooling tower to accommodate the additional HVAC loads with 20% spare capacity for future expansion, the replacement and consolidation of the existing loop pumps, the replacement of the existing heat pumps (geothermal alternate), the replacement of key rooftop HVAC components having more efficient operation and performance, and the replacement of the existing Building Automation System (BAS). Sustainable design features included the integration of a total energy, air-to-air exchange outdoor ventilation unit for the new addition, an alternate geothermal HVAC system for the entire facility, and the installation of an integrated photovoltaic cell roofing membrane for the collection of solar energy. (2011 – 2012)

Stryker Brigade Combat Team Facilities Readiness Center and Field Maintenance Shop, Easton, PA, Principal-in-Charge – Multi-disciplined engineering services for this design-build project. The new Easton Readiness Center and Field Maintenance Shop in Forks Township, Northampton County, is a unique multi-use facility that houses two Pennsylvania Army National Guard units and a field maintenance shop all under one roof. The new facility is approximately 58,000 square feet. More than 200 Army National Guard soldiers at a time train at this facility. In addition, 32 full-time employees work at the Field Maintenance Shop, servicing more than 400 vehicles annually. This project was qualified to meet the requirements of a LEED Silver building.

Gwen Howard, RA

LEED AP, Vice President of Architecture, Historic Preservation Specialist, Code Enforcement Official

Years Experience

Total Years Experience: 29
Years with Foit-Albert: 18

Education

M.Arch, School of Architecture and Planning, State University of New York at Buffalo, 1995
B.Fine Arts, Historic Preservation, Savannah College of Art and Design, 1987

Registration

Registered Architect, NY, 1998, #026733
Certified NYS Code Enforcement Official, #0697-7326B / NY0030937
LEED 2.0 Accredited Professional, 2002

Affiliations

Chairman of the Board, Buffalo Preservation Board

Chairman of the Board, YWCA of Western New York

Past Board Member, Landmark Society of the Niagara Frontier

Awards

Buffalo Zoo Entry Complex
2014 Business First Brick by Brick Award, Best Retail Project

South American Rainforest Exhibit, 2008 Decorative Concrete Award, WNY Chapter of the American Concrete Institute

Alfiero Center Addition, University of Buffalo; 2006 AIA Buffalo/WNY Chapter Honorable Mention Award

Church of Scientology Buffalo, 2004 AIA Buffalo/WNY Honor Award and 2004 Business First Brick by Brick Award, Best Historic Preservation Project

Gwen A. Howard, RA, LEED AP is Vice President of the architectural department at Foit-Albert Associates responsible for the daily management of Senior Team Leaders, Registered Architects and Technical staff. She has nearly 30 years of professional experience and has gained extensive knowledge in the areas of new-building design, rehabilitation, historic preservation, facility assessments, feasibility planning, master planning and code review.

As Sr. Project Manager, she has overseen and successfully completed projects from design through construction. Her portfolio of work includes cultural, educational, federal, municipal and historic preservation projects.

Sheriff's Department Live Shoot House

Erie County, New York

Ms. Howard served as Code Enforcement Official for the architectural drawings and specifications for a pre-engineered building to be located on the site at the Erie County Sheriff's Department shooting range site. Sr. *Project Manager*

Sheriff's Department Classroom and Tower

TRS

Ms. Howard served as Architect and Code Enforcement Official for the addition of a two-story classroom at the Erie County Sheriff's Department shooting range site. She oversaw the architectural design, code review and coordinated with the structural engineers and consultants. *Sr. Project Manager*

Lancaster Police Headquarters and Courts Facility

Town of Lancaster, NY

Ms. Howard led the team on for the completion of a new 26,000 sf building to house the Town Courts and Police Headquarters, which included court room, administrative facilities, 911 dispatch center, evidence lab, secure holding, locker rooms and tactical training firing range. She was responsible for the facility's design, code review, management of all disciplines (including subconsultants), adhering to project timetable and budget, public participation meetings and construction administration. Sr. *Project Manager*

Coxsackie/Green Correctional Facility

Coxsackie, New York

Ms. Howard was responsible for foundation and structural roof membrane design and supporting structural evaluations, as well as, code review, for a 71' by 55' classroom training building as assignment #1 under an Office of General Services Firing Range Design Term Contract. *Sr. Project Manager*

Sing Sing Correctional Facility

Ossining, NY

Ms. Howard is overseeing the architectural and interior element design for the 1st floor preparation room and the shell of the 2nd floor, and code review for the building of a new enclosed firing range. This is the second assignment of an Office of General Services Firing Range Term Contract. *Sr. Project Manager*

SHAWN M. COWE, AIA

Senior Project Manager

Years Experience

Total Years Experience: 35
Years with Foit-Albert: 16

Education

AAS, Computer Graphics,
SUNY at Alfred, 1984
AAS, Architectural Technology,
SUNY at Alfred, 1984

Registration

1994, Architecture, NY, #024433

Affiliations

American Institute of Architects

Shawn M. Cowe, AIA, is an Associate and Senior Project Manager in our architectural group. In his 35 years of professional experience he has served as project manager and designer on a variety of projects, including, residential, healthcare related facilities, educational, municipal, parks and recreation, corporate development projects, as well as commercial and retail buildings. His work involves building design, space planning, site design, construction drawings and specifications, construction contract procurement, post-occupancy evaluations, damage/defect investigations, and square footage analysis. He has also reviewed applicability and project compliance with ADA, zoning and municipal ordinances.

Lancaster Police Headquarters and Courts Facility

Town of Lancaster, NY

Mr. Cowe provided the schematic design of a new 26,000 sf building to house the Town Courts and Police Headquarters, court room, administrative facilities, 911 dispatch center, evidence lab, secure holding, locker rooms and tactical training firing range. *Project Architect*

Niagara County Community College Learning Commons

Niagara County Community College

Mr. Cowe is providing architectural design services for this renovation of former library and miscellaneous spaces into a center for collaborative learning. This project includes the new construction of a Campus Center Student Commons, as well as courtyard, laboratories, offices, childcare, gallery and limited food service areas Work included field verification of existing conditions. *Project Architect*

Frontier CSD Capital Plan

Frontier Central School District

Foit-Albert Associates provided full architectural and engineering services for this $11.3 million capital project. Work included: boiler room additional at High School, conversion of campus steam heating system to a dedicated hot water system, masonry repairs, new grandstands, press box and track and field venues, ceiling and lighting upgrades, replacement of all fire alarm systems district –wide, asbestos abatement, replacement of all flooring, new tennis courts and repair of all parking areas including drainage replacement. *Team Leader*

Morningside Estates

Ellicottville, NY

Foit-Albert Associates provided design, construction document preparation, bidding assistance and construction administration services for a 52 unit townhouse development located on Route 219 in Ellicottville, New York. Each unit includes three bedrooms, three baths, great room, loft, kitchen and one car garage. Site amenities are to include access drives, parking areas and landscaped space. *Team Leader*

Rail Station SOGR Strategic Assessment

Niagara Frontier Transportation Authority

Provided architectural support for the strategic assessment of various NFTA LRRT rail stations in Western New York as part of a term contract. Items considered as a basis for future investment encompassed building envelope, exterior pavement, interior floors, air handling systems, fire protection systems, electrical power systems, lighting, communication and plumbing systems. *Sr. Project Architect*

JASON CIURZYNSKI, PE

Sr. Structural Engineer

Years Experience

Total Years Experience: 12
Years with Foit-Albert: 12

Education

B.S., Civil Engineering,
Rochester Institute of Technology,
2009

M. Business Administration,
Canisius College, 2013

A.S., Civil Engineering Technology,
Erie County Community College,
2006

Registration

2015, Professional Engineer, NY
#095766

2014, Professional Engineer, PA
PE082413

Mr. Ciurzynski, PE, has over 12 years of experience in a multitude of engineering centered projects. He is an experienced bridge designer having designed bridges in steel, pre-stressed concrete, cast-in-place concrete and timber; has designed improvements for roadways including drainage, utility, signage, sidewalks, guide rail and landscaping enhancements; has been responsible for the inspection of various roads, bridges and culverts and has completed numerous structural inspections and retrofit projects.

Sheriff's Office Classroom and Tower

Erie County, NY

Mr. Ciurzynski was Structural Engineer for the addition of a two-story classroom at the Erie County Sheriff's Department shooting range site. He was responsible for design for a second floor classroom that will be retrofitted into an existing pre-fab building. Determined loads of new building along with 2nd floor designs to incorporate into foundation. *Project Engineer*

OGS Firing Range Design & Rehabilitation

New York State Office of General Services

Mr. Ciurzynski is providing structural engineering services as a subconsultant for the design of firing ranges and similar facilities for OGS. His first assignment is for a concrete block building with a large canopy to be used for staff training. *Structural Engineer*

University Police Building Structural Report

SUNY Alfred

Mr. Ciurzynski performed a structural assessment of the multi-story structure located at 10 Upper College Dr., Alfred, New York. The exterior, basement, first and second floors were inspected for noteworthy deficiencies. He prepared a final report with recommendations for shoring and repairs.
Sr. Structural Engineer

McMahon Hall Boilers
SUNY Alfred Satellite Boiler Installations

State University Construction Fund

Performed an analysis of the rooftop to verify the possibility for the addition of a new boiler unit at McMahon Hall. After determining the loading is not feasible, designed alternate plans at appropriate location. *Sr. Structural Engineer*

Scholes Library Chiller
SUNY Alfred Satellite Boiler Installations

State University Construction Fund

Performed an analysis of: existing structure to review the feasibility of adding new openings in the existing exterior walls for new louvers; the existing slab to determine if it can support the load of the new chiller unit in the subbasement; provided design for additional equipment pad for new chiller pump and miscellaneous wall penetrations and fireproofing needed to accommodate for new mechanical systems. *Sr. Structural Engineer*

Agriculture Science Building Boilers
SUNY Alfred Satellite Boiler Installations

State University Construction Fund

Provided structural calculations relating the to redesign of the areaway to reduce the depth and minimize the amount of excavation needed to construct new equipment. *Sr, Structural Engineer*

FRANDINA ENGINEERING AND LAND SURVEYING, PC

CIVIL ENGINEERS AND LAND SURVEYORS

NYS Certified WBE and DBE Firm

Years of Experience

35 years / 18 with FELS

Education

BS - Civil Engineering, State University of New York at Buffalo,1981;
MBA - Finance, State University of New York at Buffalo, 1983

Professional Registrations

NY #50510 Land Surveyor, 2003
NY #64412 Professional Engineer,1988
NY#35FR0983235 Real Estate Broker, 1991

Professional Affiliations

NYS Assoc. of Professional Land Surveyors
National Society of Professional Engineers
American Society of Civil Engineers
Association of Bridge Construction and Design
University of Buffalo, Dean's Advisory Board in Civil Engineering Department from 2012-present
Tau Beta Pi
Chi Epsilon

Awards:

Women of Distinction, 2018
Girl Scouts of Western New York
Meritorious Service Award, 2003, NYS Society of Professional Engineers
Engineer of the Year, 2001, SUNY at Buffalo Engineering Alumni Assoc.
Membership Development Award, 1999, NYS Society of Professional Engineers
NYS Young Engineer of the Year, 1989, NYS Society of Professsional Engineers
Young Engineer of the Year, 1989, Erie-Niagara Chapter, National Society of Professional Engineers
Basinski-Wohler Distinguished Service Award, 2003, Erie-Niagara Chapter National Society of Professional Engineers
National Young Engineer of the Year,1990, National Society of Prof. Engineers
Young Achiever, 1989, Federation of Italian- American Societies
Who's Who in Science & Engineering, 1989
Who's Who in Government, 1990

CAREER SUMMARY

Rosanne Frandina, PE, LS is the sole owner of Frandina Engineering and Land Surveying, PC. She holds a B.S. degree in Civil Engineering and an MBA, in Finance, from the University of Buffalo. She is licensed as a NYS Professional Engineer, Land Surveyor and Real Estate Broker.

She has over 35 years of experience in both the public and private sectors. She served as Director of Development under Buffalo Mayor James D. Griffin and oversaw the design and construction of many commercial projects including the Waterfront Village townhouses, the New Buffalo Industrial Park, Key Center, Rotary Rink at Fountain Plaza, the downtown overhead walkway system and many neighborhood community centers.

She served as Executive Secretary of the Buffalo Sewer Authority and then became Treatment Plant Administrator for the Bird Island Waste Water Treatment Plant. In 2000, she left public employment to join her father, Philip F. Frandina, P.E., LS at Frandina Engineering. In 2005, she established Frandina Engineering and Land Surveying as a wholly owned Woman Business Enterprise (WBE) and Disadvantaged Business Enterprise (DBE).

FELS performs Land Surveying and Civil Engineering services throughout the Western New York region including the UB Medical School, Seneca Buffalo Creek Casino, Fillmore Avenue Streetscape, Genesee Street Gateway, Niagara Falls State Park and both the Buffalo Niagara International Airport and Niagara Falls International Airport.

SELECTED PROFESSIONAL EXPERIENCE

Buffalo Bills - New Era Field Stadium Improvements (Job #3628) 1 Bills Drive, Orchard Park, NY: Land Surveying services for multiple contracts on the $130 Million stadium renovation project. Contracts included topographic surveys for design, construction stakeout for the new Team Store, 535+ light poles, all the signs at the new gates, sidewalks and ramps. Boundary and Topographic surveys for additional parcels needed for stadium and field house expansion, weight room, band shell, and stage for August 2018 Beyoncé concert.

CHA Consulting - South Ogden Bridge Replacement (Job #3991) Buffalo, NY: Topographic survey of 2000 LF of roadway and 20 hydraulic cross sections of the Buffalo River under the South Ogden Street bridge. Also included the preparation of Right of Way Acquisition maps including 15 Parcel maps with 4 Fee parcels, 7 Temporary easements and 4 Permanent easements. Contract price: $33,000.

Urban Engineers - North Canal Road (Job #4279) Lockport, NY: Topographic survey of North Canal Road from Old Niagara Road to the Erie Canal in Lockport, NY under PIN 5761.65. Included approximately 12,000 LF of roadway, utility investigation and control ties. Our contract was for $21,240.

DiDonato Associates - North and South Main Streets (Job #4406) Angola, NY: Land Surveying Services in the Village of Angola for the Erie County Department of Public Works. Our scope included a topographic survey of 16,000 LF of roadway, utility investigations and inverts and control ties. Contract price: $39,427.

Key Staff

Joseph Dommer

Executive Vice President – Senior Estimator

Background

Joe brings 27 years of industry experience to the firm. With a degree in Construction Management Technology, Mr. Dommer's experience includes many public, university, K-12, healthcare, and complex industrial projects where he has served as the Chief Cost Estimator and/or Project Manager.

Joe has supported hundreds of projects that have ranged from $100,000 to $500M in construction value. He is also a graduate of the University at Buffalo Center for Entrepreneurial Leadership. Joe's experience is rooted in his time at Baer & Associates where he started in June 1991 as a Summer intern and became a full- time employee in May 1992. Joe's career path took him through several different roles at Baer & Associates, including Quantity Estimator, Project Manager, Vice President, and President in 2004.

In 2017, he co-founded Trophy Point with Rich Chudzik and has been applying his lessons learned from the industry over the past 27 years towards growing the company. Mr. Dommer is a member of the Erie Community College Civil Engineering / Construction Management Advisory Council, the Hilbert Board of Trustees, and an affiliate member of the Buffalo-Western New York Chapter of the American Institute of Architects.

Education

- *Erie Community College, Buffalo, NY*
  Associates – Construction Management
- *University at Buffalo, Buffalo, NY*
  Core program graduate – Center for Entrepreneurial Leadership

Project Experience

- Lancaster Police and Courts
- Town of Amherst – Police Short Term Holding Addition
- Town of Clarence Police and Court
- Litigation Support – Town of Cheektowaga Police and Courts Building
- City of Buffalo Police Radio Station
- Buffalo Police Department – Female Inmate Booking at the City Court Building

Flat Hourly Fees 6

Flat hourly fees are as follows:

Title	Hourly Rates
Project Manager*	$ 117.00
QA Officer	$ 155.00
Firing Range Design Manager	$ 125.00
Sr. HVAC Engineer / Acoustical Designer	$ 130.00
Sr. Mechanical Electrical Engineer	$ 125.00
Jr. Mechanical Electrical Engineer	$ 93.00
Sr. Project Architect	$ 130.00
Project Architect	$ 95.00
Sr. Structural Engineer	$ 130.00
Sr. Environmental Geologist*	$ 105.00
Environmental Scientist*	$ 82.00
CAD Technician*	$ 62.00

Rates are inclusive of all overhead and fees. LiRo’s Contract Executive will not bill the project.

* Environmental Phase Rates

Subconsultants and Encouraging Disadvantaged Group Participation 7

Subconsultants

LiRo has developed a team that includes Minority- and Woman-Owned Businesses Enterprises (M/WBE) with whom we have excellent routine and complementary working relationships. We have partnered with these firms not because of their status, but because of their shared commitment and track record of meeting our Clients' expectations and their enhancement of our team with their particular areas of expertise. Our subcontracted partners include:

- Foit-Albert Associates (MBE) – Architecture/Structural Engineering
- Frandina Engineering and Land Surveying, PC (WBE) – Locational Surveying
- Nature's Way Environmental (WBE) – Boring and Geotechnical Services
- Trophy Point Construction Services and Consulting (SDVOB) – Cost Estimating

In addition, we will contract with WBE or MBE laboratories for the environmental and material testing project needs, once these particular requirements are known. Our team will exceed the City's participation goals for the project as shown below:

Status	Goal	LiRo Budget
MBE	Greater than 25 percent	Greater than 30 percent
WBE	Greater than 5 percent	Greater than 5 percent

Although not a City goal for this project, LiRo will subcontract with a Service Disabled Veteran Owned Business, Trophy Point Construction Services and Consulting, to support the team by performing Third Party Certified Cost Estimates for the Design Development and Construction Document phases of the project. This firm's participation will exceed 5-percent of our budget.

Our key subcontracted team members include:

Foit-Albert Associates is a Buffalo based award-winning multidisciplinary architecture, engineering and surveying consulting firm. The architectural group brings extensive experience in the design of new buildings and the rehabilitation and restoration of existing facilities. Clients include public and private sectors, and encompass federal, state, municipal, commercial, institutional and industrial projects. Project types include colleges and universities, K-12 educational institutions, municipal facilities, cultural and heritage facilities, housing projects, zoos and aquariums, healthcare institutions, parks & recreation and shooting range projects. The firm also has historic preservation specialists and a certified code-enforcement official.

Frandina Engineering and Land Surveying, PC provides high quality land surveying services throughout Western New York. In 2005, Rosanne Frandina, PE, LS, established Frandina Engineering and Land Surveying, PC as a wholly-owned Woman Business Enterprise. The firm is also a certified Disadvantaged Business Enterprise. Her staff has significant experience working on the largest jobs in the Western New York region. Their field work and mapping is trusted by engineers, architects and developers to provide the basis of design for building these high-profile projects.

Trophy Point is a leader in the construction services industry and the largest independent source of construction cost estimates in Western New York. They are dedicated to providing accurate, timely and thorough estimates and reports. They have worked on more than 5,000 projects throughout the Northeast and understand a wide range of cost analysis and reporting requirements. Trophy Point has a proven track record for providing thorough budgets and accurate bid-day projections, providing their clients peace-of-mind. Reports produced by Trophy Point are designed with the end user in mind and offer a clear narrative of project scope along with quantified costs.

Nature's Way Environmental, Consultants and Contractors is a multi-service drilling company specializing in drilling (environmental services and geotechnical services) and direct push services for a multitude of projects. Based in Alden, NY, a suburb of Buffalo, Nature's Way Environmental is a family-owned company. Nature's Way was founded in 1989 as a petroleum remediation contractor specializing in bio-remediation. Since then, their services have grown to include tank removals, excavation/disposal of contaminated soil and groundwater, complete in-situ remediation systems, environmental and geotechnical subsurface investigations/drilling, and complete fuel system installs and maintenance. They hold certifications as a women-owned business entity (WBE) in the City of Buffalo/Erie County and in New York State**.**

Encouraging Female and Minority Participation

LiRo makes every effort to hire qualified, able employees and to give everyone an equal opportunity to progress within the Company. LiRo is firmly committed to the principles of equal employment opportunity and to compliance with all federal, state and local laws concerning employment discrimination. To this end, LiRo ensures equal employment opportunity to all employees and applicants regardless of race, color, creed, age, sex, sexual orientation, religion, marital status, national origin or ancestry, citizenship, physical or mental handicap, disability, veteran status or liability for service in the U.S. Armed Forces or any other status or condition protected by applicable federal, state or local laws. The policy of equal opportunity will be observed with respect to all employment practices including, without limitation, recruitment, hiring (or failure or refusal to hire), training and development, job assignment, tenure, promotion, layoff, transfer, termination, compensation, benefits, working conditions, and other obligations and privileges of employment in accordance with applicable federal, state and local laws.

Mentoring Women Owned and Minority Owned Small Businesses

The LiRo Group has a Training and Mentoring Program that provides support at all stages of projects (from Program Inception to Closeout). The components address the need to provide small contractors with access to training, administrative tools and mentoring needed to become a successful contractor.

The goal is to design and implement a comprehensive program that prepares contractors and professional services firms in various aspects of a construction program. The goal is to develop and implement training and mentoring components that address the following themes such as The Construction Business and General Business Acumen, Construction Projects and Contracts and Managing Projects, and People and Materials.

In addition, LiRo has also conducted on-site training to assist firms with preparing invoices and supporting documentation. Working together with our network of M/WBE's, and collaborators such as the Governor's Office of Storm Recovery, we have compiled and developed substantial resources to organize and manage not only a successful M/WBE program but one that also employs local, qualified talent that promotes self-sufficiency.